

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Syngas compressor Antisurge cooler-2-AK.pvdb

Date of Analysis : Sep 7, 2026 2:56pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1510.000 mm ID	
Vessel Design Length, Tangent to Tangent	1047.14 cm
Specified Datum Line Distance	0.00 cm
Shell Side Design Temperature	93 °C
Channel Side Design Temperature	85 °C
Shell Side Design Pressure	194.770 kgf/cm ²
Channel Side Design Pressure	149.902 kgf/cm ²
Shop Shell Side Test Pressure	248.441 kgf/cm ²
Shop Channel Side Test Pressure	189.642 kgf/cm ²
Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Flange		...	...		
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes		...	...		
Tubesheet	SA-266 2	...	...	K03506	+IT
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind Flg	150.167	0.00	152.000	3.0000	N/A	No
LHS Ch Flg	150.053	0.00	155.000	3.0000	N/A	No
LHS Ch Shell	150.053	0.00	156.000	3.0000	N/A	No
Main Shell	194.778	0.00	199.000	3.0000	N/A	No
RHS Ch Shell	150.053	0.00	153.000	3.0000	N/A	No
RHS Ch Flg	150.053	0.00	153.000	3.0000	N/A	No
RHS Blind Flg	150.166	0.00	152.000	3.0000	N/A	No

Liquid Level: 265.00 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	45.8	45.5	463.0	455.0	452.3	...	1.00	1.00
Body Flg	107.8	62.0	448.0	440.0	436.9	...	1.00	1.00
Cylinder	167.8	60.0	92.0	92.0	88.3	...	1.00	1.00
Cylinder	871.3	675.5	118.0	118.0	115.5	...	1.00	1.00
Cylinder	939.3	40.0	90.0	90.0	88.3	...	1.00	1.00
Body Flg	1001.6	62.0	448.0	440.0	435.2	...	1.00	1.00

Body Flg	1047.1	45.5	463.0	455.0	452.3	...	1.00	1.00
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Local Stress Analysis Results:

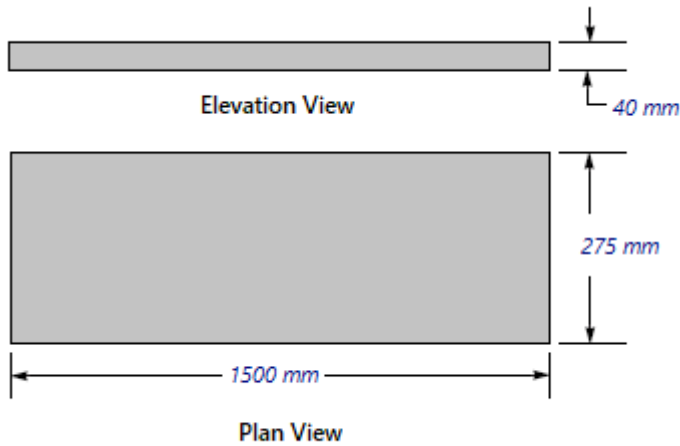
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
T2-Tube outlet	WRC-297	0.963	Shell	Passed
T1-Tube Inlet	WRC-297	0.964	Shell	Passed
S1-Shell Inlet	WRC-297	0.986	Shell	Passed
S2-Shell outle	WRC-297	0.986	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	250.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1300.000	mm
Wear Pad Width	740.000	mm
Wear Pad Thickness	24.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	755.000	mm
Baseplate Length	1500.000	mm
Baseplate Thickness	40.000	mm
Baseplate Width	275.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	24.000	mm
Web Thickness	24.000	mm
Height of Center Web	500.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	121946.30	kgf
Maximum Transverse Saddle Shear Load	8927.47	kgf
Maximum Longitudinal Saddle Shear Load	17854.94	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	100474.84	kgf
Maximum Transverse Saddle Shear Load	331.64	kgf
Maximum Longitudinal Saddle Shear Load	202.56	kgf

Weights:

Fabricated - Bare w/o removable internals	178317.5	kg
Shop Test - Fabricated + water (full)	192972.5	kg
Shipping - Fab. + rem. intls.+ shipping app.	179069.7	kg



Erected	- Fab. + rem. intls.+ insul. (etc)	179069.7	kg
Empty	- Fab. + intls. + details + wghts.	179069.7	kg
Operating	- Empty + operating liquid (no ca)	185069.8	kg
Field Test	- Empty weight + test medium (full)	179565.6	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
T2-Tube outlet	156	...	...	OK	OK	Passed
T1-Tube Inlet	156	...	...	OK	OK	Passed
S1-Shell Inlet	199	...	...	OK	OK	Passed
S2-Shell outlet	199	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 199 Nozzle : S2-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 156 Nozzle : T1-Tube Inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

- Error: Nozzle T2-Tube outlet, was too close to a gross structural discontinuity.
- Error: Nozzle T1-Tube Inlet, was too close to a gross structural discontinuity.

Please see ASME VIII-2 Section 4.5.13.1 for more information.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
30	T2-Tube outlet	1378.175	90.000	130.000
30	T1-Tube Inlet	1378.175	270.000	130.000
40	S1-Shell Inlet	3458.175	90.000	255.425
40	S2-Shell outlet	6958.174	270.000	255.425

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
T2-Tube outlet	10.000 in	Actual	RTJWn	15.512	70.000	...	...	426.93	1500
T1-Tube Inlet	10.000 in	Actual	RTJWn	15.512	70.000	...	...	426.93	1500
S1-Shell Inlet	24.000 in	140	RTJWn	24.000	52.375	...	...	568.57	1500
S2-Shell outlet	24.000 in	140	RTJWn	24.000	52.375	...	...	568.57	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
T2-Tube out	SA-266 2	90.000	10.000	...	...	...
T1-Tube Inl	SA-266 2	90.000	10.000	...	...	...
S1-Shell In	SA-266 2	118.000	10.000	...	...	...
S2-Shell ou	SA-266 2	118.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
T2-Tube outlet	137.818	90.0	300.00	0.00	LHS Ch Shell
T1-Tube Inlet	137.818	270.0	300.00	0.00	LHS Ch Shell
S1-Shell Inlet	345.817	90.0	300.00	0.00	Main Shell
S2-Shell outlet	695.817	270.0	300.00	0.00	Main Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 463.0mm THK X 2650.0mm OD	SA-266 2
2	WELD NECK FLANGE: 448.0mm THK X 2645.0mm OD	SA-266 2
1	CYLINDER: 92.0mm THK X 1510.0mm ID X 60.0cm	SA-516 70
1	BLIND FLANGE: 463.0mm THK X 2645.0mm OD	SA-266 2
7	BAFFLES: 1450.0mm DIA X 8.0mm THK	SA-516 70
8	TIE RODS: 19mm DIA X 6755mm	SA-193 B7M
2	TUBESHEET: 1746.0mm X 280.0mm THK	SA-266 2
1548	TUBES: 7315.0mm X 19.0mm DIA X 3.0mm THK	SA-179
2	SADDLE: 250.0mm X 120 DEG	SA-516 70
1	INSULATION: 6755mm X 100mm THK	Hot Insulation
2	GASKET: 1574.0mm OD X 1542.0mm ID	...
96	BODY FLANGE BOLTS: 80.0mm DIA	SA-193 B7
192	NUTS FOR BODY FLANGE BOLTS: 80.0mm DIA	...
1	CYLINDRICAL SEGMENT 118.0mm THK X 3048.0mm X 675.5cm	SA-516 70
1	CYLINDRICAL SEGMENT 118.0mm THK X 1695.8mm X 675.5cm	SA-516 70
1	CYLINDRICAL SEGMENT 90.0mm THK X 3048.0mm X 40.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 90.0mm THK X 1695.8mm X 40.0cm	SA-516 70
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Main Shell [8]	!	-10			Yes			
S1-Shell Inlet[1]	B	28	23		Yes	0.885	45.828	1.00
Nozzle Flg [5]	A	-18	-32		Yes			
S2-Shell outle[1]	B	28	23		Yes	0.885	45.828	1.00
Nozzle Flg [5]	A	-18	-32		Yes			
Nozl. Bolting[22]		-48			No	1.000	64.000	1.00
Tubesheet: SS[13]	!	-18			Yes			
Warmest MDMT:		28	23					
LHS Blind Flg[11][17] Impact Testing reqd					No			
LHS Ch Flg [11]	B	20	19		Yes	0.971	92.000	1.00
RHS Ch Flg [11]	B	20	19		Yes	0.981	90.000	1.00
RHS Blind Flg[11][17] Impact Testing reqd					No			
LHS Ch Shell [8]	!	0			Yes			
RHS Ch Shell [8]	!	0			Yes			
T2-Tube outlet[1]	B	20	18		Yes	0.957	92.000	1.00
Nozzle Flg [5]	A	-18	-42		Yes			
T1-Tube Inlet [1]	B	20	18		Yes	0.958	92.000	1.00
Nozzle Flg [5]	A	-18	-42		Yes			
Nozl. Bolting[22]		-48			No	1.000	64.000	1.00
Tubesheet: CS[14]	!	-18			Yes			
Bolting [21]		-40						
Bolting [21]		-40						
Warmest MDMT:		20	19					
Exchanger Side		Computed MDMT		Required MDMT		Pass/Fail		
Shell		23.0		-10.0		Fail		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.

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Units used in this Analysis (Kgfc/m2):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	194.77	kgf/cm ²
Channel Side Design Pressure	149.9	kgf/cm ²
Shell Side Design Temperature	93.3	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-10.0	°C
MDMT (CET), Tube Side	0.0	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	Pneumatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 cm
Distance Upwind of Crest	Lh 0 cm
Distance from Crest to the Vessel	x 0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
--	-----



Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Blind Flg
Distance "FROM" to "TO"	45.5 cm
Flange Outside Diameter	2650 mm
Element Thickness	455 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	463 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	149.9 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2294.2 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2340.9 kgf/cm ²
MDMT Curve	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	265 cm
Liquid Density	999.54 kg/m ³

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	LHS Ch Flg
Distance "FROM" to "TO"	62 cm
Flange Inside Diameter	1510 mm
Element Thickness	440 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	448 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	149.9 kgf/cm ²
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid



Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	151	cm
Liquid Density	999.54	kg/m ³

Element#	3/7	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	LHS Ch Shell	
Distance "FROM" to "TO"	60	cm
Inside Diameter	1510	mm
Element Thickness	92	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	92	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	149.9	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2467.6	kgf/cm ²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	151	cm
Liquid Density	999.54	kg/m ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	T2-Tube outlet	
Dist. from "FROM" Node / Offset dist	30	cm
Nozzle Diameter	10	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	495.24	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	T1-Tube Inlet	
Dist. from "FROM" Node / Offset dist	30	cm
Nozzle Diameter	10	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	495.24	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	



Element From Node	30
Detail Type	Weight
Detail ID	Datasheet weigh
Dist. from "FROM" Node / Offset dist	0 cm
Miscellaneous Weight	1000 kgf
Offset from Element Centerline	0 mm

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	675.5 cm
Inside Diameter	1510 mm
Element Thickness	118 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	118 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	194.77 kgf/cm ²
Design Temperature Internal Pressure	93 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	75.5 cm
Width of Saddle	250 mm
Height of Saddle at Bottom	1300 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	740 mm
Thickness of Wear Plate	24 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 cm

Element From Node	40
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	600 cm
Width of Saddle	250 mm
Height of Saddle at Bottom	1300 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	740 mm
Thickness of Wear Plate	24 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	7.9999998e-002
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 cm

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40



Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	151	cm
Liquid Density	50.177	kg/m ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	675.5	cm
Thickness of Insulation	100	mm
Density	0.0001918	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	S1-Shell Inlet	
Dist. from "FROM" Node / Offset dist	150	cm
Nozzle Diameter	24	in.
Nozzle Schedule	140	
Nozzle Class	1500	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2451.8	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	S2-Shell outlet	
Dist. from "FROM" Node / Offset dist	500	cm
Nozzle Diameter	24	in.
Nozzle Schedule	140	
Nozzle Class	1500	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2577.3	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	40	
Detail Type	Weight	
Detail ID	margin	
Dist. from "FROM" Node / Offset dist	350	cm
Miscellaneous Weight	9999.8	kgf
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	datasheet weigh	
Dist. from "FROM" Node / Offset dist	350	cm
Miscellaneous Weight	34999	kgf
Offset from Element Centerline	0	mm

Element#	5/7	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	RHS Ch Shell	
Distance "FROM" to "TO"	40	cm
Inside Diameter	1510	mm
Element Thickness	90	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	90	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	149.9	kgf/cm ²
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	



Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Liquid
 Detail ID Liquid: 50
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 151 cm
 Liquid Density 999.54 kg/m³

Element# 6/7
 Element From Node 60
 Element To Node 70
 Element Type Flange
 Description RHS Ch Flg
 Distance "FROM" to "TO" 62 cm
 Flange Inside Diameter 1510 mm
 Element Thickness 440 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 448 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 149.9 kgf/cm²
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-266 2
 Allowable Stress, Ambient 1407.1 kgf/cm²
 Allowable Stress, Operating 1407.1 kgf/cm²
 Allowable Stress, Hydrotest 2294.2 kgf/cm²
 Material Density 0.00775 kg/cm³
 P Number Thickness 32.004 mm
 Yield Stress, Operating 2340.9 kgf/cm²
 MDMT Curve B
 External Pressure Chart Name CS-2
 UNS Number K03506
 Product Form Forgings
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element From Node 60
 Detail Type Liquid
 Detail ID Liquid: 60
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 151 cm
 Liquid Density 999.54 kg/m³

Element# 7/7
 Element From Node 70
 Element To Node 80
 Element Type Flange
 Description RHS Blind Flg
 Distance "FROM" to "TO" 45.5 cm
 Flange Outside Diameter 2645 mm
 Element Thickness 455 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 463 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 149.9 kgf/cm²
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-266 2



Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	264.5 cm
Liquid Density	999.54 kg/m ³

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XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
LHS Blind Flg		45.8175	...	45.5	...
LHS Ch Flg		107.818	...	62	...
LHS Ch Shell		167.818	...	60	...
Main Shell		871.318	...	675.5	...
RHS Ch Shell		939.318	...	40	...
RHS Ch Flg		1001.64	...	62	...
RHS Blind Flg		1047.14	...	45.5	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

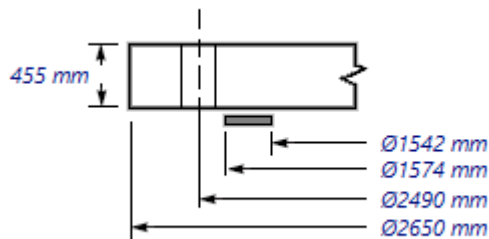
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Flange Input Data Values

Description: LHS Blind Fl :

LHS Blind Flg

Flange Type	TEMA Channel Cover		
Design Pressure	P	150.17	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2650.000	mm
Flange Thickness	t	455.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²
Perimeter along the Center of the Bolts	L	7822.566	mm
Youngs Modulus for Blind Flange Matl.	Sy	2029713.75	kgf/cm ²
Allowed Channel Cover Deflection	dfAllow	1.8875	mm
Diameter of Bolt Circle	C	2490.000	mm
Nominal Bolt Diameter	a	80.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		48	
Flange Face Outside Diameter	Fod	1585.000	mm
Flange Face Inside Diameter	Fid	1510.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1574.000	mm
Gasket Inside Diameter	Gi	1542.000	mm
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	175.77	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	16.000 mm
Basic Gasket Width,	$bo = N/2$	8.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1559.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1560^2 \cdot 150$

= 2869262.000 kgf

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1560 \cdot 2 \cdot 150$$

$$= 209782.016 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(2869262 + 209782 + 0, 0)$$

$$= 3079044.000 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 176 \cdot 7.13 \cdot \pi \cdot 1560 + 0 \cdot 0 \cdot 0$$

$$= 61386.547 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((3079044 + 0 + 4 \cdot 0 / 1560) / 1621, 61387 / 1621)$$

$$= 1899.166 \text{ cm}^2$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1899.166	1956.879	
Radial Distance between Bolts and Edge:	74.000	80.000	
Circ. Spacing between the Bolts:	162.600	162.854	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1957 \cdot 1621 / (176 \cdot \pi (1574 + 1542))$$

$$= 184.388 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 1957$$

$$= 3172611.75 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 3079044.00 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2490 - 1560) / 2$$

$$= 465.1274 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 3079044 \cdot 465 \cdot 1 / (452^2 \cdot 1560) + 0.3 \cdot 1 \cdot 150 \cdot 1560^2 / 452^2$$

$$= 1390.35 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 3172611 \cdot 465 \cdot 1 / (452^2 \cdot 1560)$$

$$= 879.86 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 3079044 / 1957$$

$$= 1573.45 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:



= (Wm2 / Ab)
 = (61387/1957)
 = 31.37 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	1390.35	1407.13	879.86	1407.13
Bolting	1573.45	1621.26	31.37	1621.26

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 0.8203 mm
 Allowed deflection at center of Cover 1.8875 mm

Reqd. Blind Flange Thickness at Center 452.297 mm
 Reqd. Blind Flange Thickness at Gasket 360.418 mm
 Min. Required Flange Thickness per TEMA 342.367 mm

Estimated M.A.W.P. (Operating) 152 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 19449.8 kg
 Estimated Unfinished Weight of Forging at given Thk 19449.8 kg

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	1372.08	1407.13	868.29	1407.13
Bolting	1573.45	1621.26	31.37	1621.26

Minimum Req'd Thickness, ambient, no Corrosion: 449.301 mm

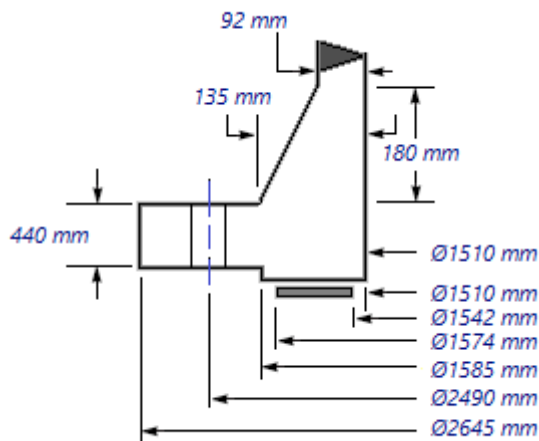
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Flange Input Data Values

Description: LHS Ch Flg :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	150.05	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1510.000	mm
Flange Outside Diameter	A	2645.000	mm
Flange Thickness	t	440.0000	mm
Thickness of Hub at Small End	go	92.0000	mm
Thickness of Hub at Large End	gl	135.0000	mm
Length of Hub	h	180.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²
Diameter of Bolt Circle	C	2490.000	mm
Nominal Bolt Diameter	a	80.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		48	
Flange Face Outside Diameter	Fod	1585.000	mm
Flange Face Inside Diameter	Fid	1510.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1574.000	mm
Gasket Inside Diameter	Gi	1542.000	mm
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	175.77	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (150(1510+3)/(1407-0.5 \cdot 150))+(3)$$

= 88.3838 mm

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 1407 \cdot \log((2 \cdot 89 + 758) / 758)$$

$$= 156.216 \text{ kgf/cm}^2$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2.0 \cdot T + R) / R)$$

$$= 1407 \cdot \log((2.0 \cdot 92 + 755) / 755)$$

$$= 161.796 \text{ kgf/cm}^2$$

Corroded Flange Thickness, tc = T-ci	437.000 mm
Corroded Flange ID, Bcor = B+2*ci	1516.000 mm
Corroded Large Hub, g1Cor = g1-ci	132.000 mm
Corroded Small Hub, g0Cor = go-ci	89.000 mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	355.000 mm

Gasket Contact Width, N = (Go - Gi)/2	16.000 mm
Basic Gasket Width, bo = N/2	8.000 mm
Effective Gasket Width, b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter, G = Go - 2 * b	1559.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1560^2 \cdot 150$$

$$= 2867084.500 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1560 \cdot 2 \cdot 150$$

$$= 209622.828 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1516^2 \cdot 150$$

$$= 2708517.250 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 2867084 - 2708517$$

$$= 158567.141 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(2867084 + 209623 + 0, 0)$$

$$= 3076707.250 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 176 \cdot 7.13 \cdot \pi \cdot 1560 + 0 \cdot 0 \cdot 0$$

$$= 61386.555 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((3076707 + 0 + 4 \cdot 0 / 1560) / 1621, 61387 / 1621)$$

$$= 1897.725 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 80 + 6 \cdot 437 / (2 + 0.5)$$

$$= 1208.800 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2490 \cdot \sin(\pi / 48)$$

$$= 162.854 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{163/(2 \cdot 80 + 437)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1897.725	1956.879	
Radial Distance between Hub and Bolts:	91.500	355.000	
Radial Distance between Bolts and Edge:	74.000	77.500	
Circ. Spacing between the Bolts:	162.600	162.854	1208.800

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 1957 \cdot 1621 / (176 \cdot \pi(1574 + 1542))$$

$$= 184.388 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 1957$$

$$= 3172611.75 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 3076707 - 2867084$$

$$= 209622.78 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2490 - 1560) / 2$$

$$= 465.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2490 - 1516) / 2 + 465)$$

$$= 476.0637 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2490 - 1516 - 132) / 2$$

$$= 421.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	2708517.	421.0000	1.0000	1140287.
Face Pressure, Mt	158567.	476.0637	1.0000	75488.
Gasket Load, Mg	209623.	465.1274	1.0000	97501.
Gasket Seating, Matm	3172612.	465.1274	1.0000	1475670.

Total Moment for Operation, Mo 1313277. kgf-m.
 Total Moment for Gasket seating, Mg 1475670. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 367.320 mm
 Hub Ratio, h/h0 = HL / H0 0.490
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.483

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.850
 Factor V per 4.16.5 0.318

Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2645/1516$$

$$= 1.745$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.74^2 \cdot \log_{10}(1.74))) / (1.74^2 - 1)) / (1.74 - 1)$$

$$= 3.661$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.74^2(1 + 8.55246 \cdot \log_{10}(1.74)) - 1) / ((1.0472 + 1.9448 \cdot 1.74^2) \cdot (1.74 - 1))$$

$$= 1.607$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.74^2(1 + 8.55246 \cdot \log_{10}(1.74)) - 1) / (1.36136(1.74^2 - 1) \cdot (1.74 - 1))$$

$$= 4.023$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.74^2 + 1) / (1.74^2 - 1)$$

$$= 1.978$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.02 \cdot 89^2 \cdot 367 / 0.32$$

$$= 36757404.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.85/367$$

$$= 0.002 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (437 \cdot 0.0023 + 1) / 1.61 + 437^3/36757404$$

$$= 3.522$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 1313276/1516) / (3.52 \cdot 132^2)$$

$$= 1411.73 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 1475670/1516) / (3.52 \cdot 132^2)$$

$$= 1586.30 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.35 \cdot 1313276/1516) / (3.52 \cdot 437^2)$$

$$= 302.38 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.35 \cdot 1475670/1516) / (3.52 \cdot 437^2)$$

$$= 339.77 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.66 \cdot 1313276 / (437^2 \cdot 1516)) - 1.98 \cdot 302$$

$$= 1062.47 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (3.66 \cdot 1475670 / (437^2 \cdot 1516)) - 1.98 \cdot 340$$

$$= 1193.85 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (1412 + \max(302, 1062)) / 2$$

$$= 1237.10 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (1586 + \max(340, 1194)) / 2$$

$$= 1390.08 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 3076707 / 1957$$

$$= 1572.25 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (61387 / 1957)$$

$$= 31.37 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	1411.73	2110.70	1586.30	2110.70
Radial Flange	302.38	1407.13	339.77	1407.13
Tangential Flange	1062.47	1407.13	1193.85	1407.13
Maximum Average	1237.10	1407.13	1390.08	1407.13
Bolting	1572.25	1621.26	31.37	1621.26

Minimum Required Flange Thickness 436.880 mm

Estimated M.A.W.P. (Operating) 155 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 13438.4 kg
 Estimated Unfinished Weight of Forging at given Thk 17798.0 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1475670 / 1 \cdot 100000 \cdot 0.32 / (3.47 \cdot 2067025 \cdot 89^2 \cdot 367 \cdot 0.3)$$

$$= 0.391 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1313276 / 1 \cdot 100000 \cdot 0.32 / (3.47 \cdot 2029713 \cdot 89^2 \cdot 367 \cdot 0.3)$$

$$= 0.354 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1475670 / 1 \cdot 100000 \cdot 0.32 / (3.52 \cdot 2067025 \cdot 89^2 \cdot 367 \cdot 0.3)$$

$$= 0.385 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1313276 / 1 \cdot 100000 \cdot 0.32 / (3.52 \cdot 2029713 \cdot 89^2 \cdot 367 \cdot 0.3)$$

$$= 0.349 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 19 °C
 User entered Minimum Design Temperature : 0 °C
 The Curve for this Material is : B



From Figure 3.8M- with PWHT, the MDMT is : 20 °C
 Temperature reduction per Graph 3.13 : 1 °C
 The computed stress ratio is : 0.971

Warning: Impact Testing Is Mandatory!

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1391.90	2110.70	1557.52	2110.70
Radial Flange	306.69	1407.13	343.18	1407.13
Tangential Flange	1037.06	1407.13	1160.45	1407.13
Maximum Average	1214.48	1407.13	1358.99	1407.13
Bolting	1572.25	1621.26	31.37	1621.26

Minimum Req'd Thickness, ambient, no Corrosion: 430.936 mm

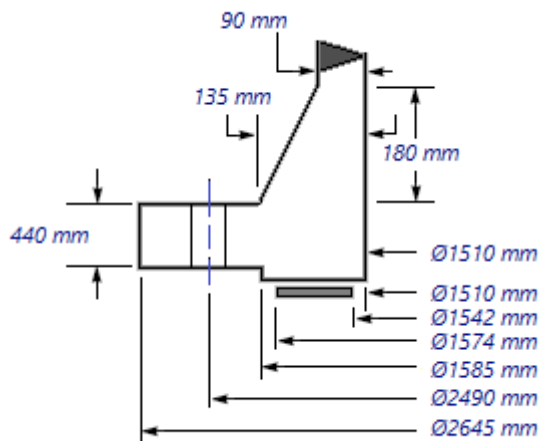
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Flange Input Data Values

Description: RHS Ch Flg :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	150.05	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1510.000	mm
Flange Outside Diameter	A	2645.000	mm
Flange Thickness	t	440.0000	mm
Thickness of Hub at Small End	go	90.0000	mm
Thickness of Hub at Large End	gl	135.0000	mm
Length of Hub	h	180.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²
Diameter of Bolt Circle	C	2490.000	mm
Nominal Bolt Diameter	a	80.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		48	
Flange Face Outside Diameter	Fod	1585.000	mm
Flange Face Inside Diameter	Fid	1510.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1574.000	mm
Gasket Inside Diameter	Gi	1542.000	mm
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	175.77	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (150(1510+3) / (1407-0.5 \cdot 150)) + (3)$$

= 88.3838 mm

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 1407 \cdot \log((2 \cdot 87 + 758) / 758)$$

$$= 152.889 \text{ kgf/cm}^2$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2.0 \cdot T + R) / R)$$

$$= 1407 \cdot \log((2.0 \cdot 90 + 755) / 755)$$

$$= 158.469 \text{ kgf/cm}^2$$

Corroded Flange Thickness, tc = T-ci	437.000 mm
Corroded Flange ID, Bcor = B+2*ci	1516.000 mm
Corroded Large Hub, g1Cor = g1-ci	132.000 mm
Corroded Small Hub, g0Cor = go-ci	87.000 mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	355.000 mm

Gasket Contact Width, N = (Go - Gi)/2	16.000 mm
Basic Gasket Width, bo = N/2	8.000 mm
Effective Gasket Width, b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter, G = Go - 2 * b	1559.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1560^2 \cdot 150$$

$$= 2867084.500 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1560 \cdot 2 \cdot 150$$

$$= 209622.828 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1516^2 \cdot 150$$

$$= 2708517.250 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 2867084 - 2708517$$

$$= 158567.141 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(2867084 + 209623 + 0, 0)$$

$$= 3076707.250 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 176 \cdot 7.13 \cdot \pi \cdot 1560 + 0 \cdot 0 \cdot 0$$

$$= 61386.547 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((3076707 + 0 + 4 \cdot 0 / 1560) / 1621, 61387 / 1621)$$

$$= 1897.725 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 80 + 6 \cdot 437 / (2 + 0.5)$$

$$= 1208.800 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2490 \cdot \sin(\pi / 48)$$

$$= 162.854 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{163/(2 \cdot 80 + 437)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1897.725	1956.879	
Radial Distance between Hub and Bolts:	91.500	355.000	
Radial Distance between Bolts and Edge:	74.000	77.500	
Circ. Spacing between the Bolts:	162.600	162.854	1208.800

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 1957 \cdot 1621 / (176 \cdot \pi(1574 + 1542))$$

$$= 184.388 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 1957$$

$$= 3172611.75 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 3076707 - 2867084$$

$$= 209622.78 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2490 - 1560) / 2$$

$$= 465.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2490 - 1516) / 2 + 465)$$

$$= 476.0637 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2490 - 1516 - 132) / 2$$

$$= 421.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	2708517.	421.0000	1.0000	1140287.
Face Pressure, Mt	158567.	476.0637	1.0000	75488.
Gasket Load, Mg	209623.	465.1274	1.0000	97501.
Gasket Seating, Matm	3172612.	465.1274	1.0000	1475670.

Total Moment for Operation, Mo 1313277. kgf-m.
 Total Moment for Gasket seating, Mg 1475670. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 363.169 mm
 Hub Ratio, h/h0 = HL / H0 0.496
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.517

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.846
 Factor V per 4.16.5 0.308

Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2645/1516$$

$$= 1.745$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.74^2 \cdot \log_{10}(1.74))) / (1.74^2 - 1)) / (1.74 - 1)$$

$$= 3.661$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.74^2(1 + 8.55246 \cdot \log_{10}(1.74)) - 1) / ((1.0472 + 1.9448 \cdot 1.74^2) \cdot (1.74 - 1))$$

$$= 1.607$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.74^2(1 + 8.55246 \cdot \log_{10}(1.74)) - 1) / (1.36136(1.74^2 - 1) \cdot (1.74 - 1))$$

$$= 4.023$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.74^2 + 1) / (1.74^2 - 1)$$

$$= 1.978$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.02 \cdot 87^2 \cdot 363 / 0.31$$

$$= 35854768.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.85/363$$

$$= 0.002 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (437 \cdot 0.0023 + 1) / 1.61 + 437^3/35854768$$

$$= 3.584$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 1313276/1516) / (3.58 \cdot 132^2)$$

$$= 1387.32 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 1475670/1516) / (3.58 \cdot 132^2)$$

$$= 1558.87 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.36 \cdot 1313276/1516) / (3.58 \cdot 437^2)$$

$$= 298.46 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.36 \cdot 1475670/1516) / (3.58 \cdot 437^2)$$

$$= 335.36 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.66 \cdot 1313276 / (437^2 \cdot 1516)) - 1.98 \cdot 298$$

$$= 1070.24 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3.66 \cdot 1475670 / (437^2 \cdot 1516)) - 1.98 \cdot 335$$

$$= 1202.58 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (1387 + \max(298, 1070)) / 2$$

$$= 1228.78 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (1559 + \max(335, 1203)) / 2$$

$$= 1380.73 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 3076707 / 1957$$

$$= 1572.25 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (61387 / 1957)$$

$$= 31.37 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	1387.32	2110.70	1558.87	2110.70
Radial Flange	298.46	1407.13	335.36	1407.13
Tangential Flange	1070.24	1407.13	1202.58	1407.13
Maximum Average	1228.78	1407.13	1380.73	1407.13
Bolting	1572.25	1621.26	31.37	1621.26

Minimum Required Flange Thickness 435.153 mm

Estimated M.A.W.P. (Operating) 153 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 13430.8 kg
 Estimated Unfinished Weight of Forging at given Thk 17798.0 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1475670 / 1 \cdot 100000 \cdot 0.31 / (3.5 \cdot 2067025 \cdot 87^2 \cdot 363 \cdot 0.3)$$

$$= 0.397 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1313276 / 1 \cdot 100000 \cdot 0.31 / (3.5 \cdot 2029713 \cdot 87^2 \cdot 363 \cdot 0.3)$$

$$= 0.360 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1475670 / 1 \cdot 100000 \cdot 0.31 / (3.58 \cdot 2067025 \cdot 87^2 \cdot 363 \cdot 0.3)$$

$$= 0.388 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1313276 / 1 \cdot 100000 \cdot 0.31 / (3.58 \cdot 2029713 \cdot 87^2 \cdot 363 \cdot 0.3)$$

$$= 0.352 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 19 °C
 User entered Minimum Design Temperature : 0 °C
 The Curve for this Material is : B



From Figure 3.8M- with PWHT, the MDMT is : 20 °C
 Temperature reduction per Graph 3.13 : 1 °C
 The computed stress ratio is : 0.981

Warning: Impact Testing Is Mandatory!

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed	
Longitudinal Hub	1368.58	2110.70	1531.43	2110.70
Radial Flange	302.83	1407.13	338.86	1407.13
Tangential Flange	1044.64	1407.13	1168.94	1407.13
Maximum Average	1206.61	1407.13	1350.19	1407.13
Bolting	1572.25	1621.26	31.37	1621.26

Minimum Req'd Thickness, ambient, no Corrosion: 429.260 mm

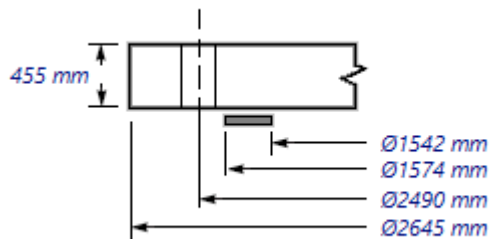
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Flange Input Data Values

Description: RHS Blind Fl :

RHS Blind Flg

Flange Type	TEMA Channel Cover		
Design Pressure	P	150.17	kgf/cm ²
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2645.000	mm
Flange Thickness	t	455.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²
Perimeter along the Center of the Bolts	L	7822.566	mm
Youngs Modulus for Blind Flange Matl.	Sy	2029713.75	kgf/cm ²
Allowed Channel Cover Deflection	dfAllow	1.8875	mm
Diameter of Bolt Circle	C	2490.000	mm
Nominal Bolt Diameter	a	80.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		48	
Flange Face Outside Diameter	Fod	1585.000	mm
Flange Face Inside Diameter	Fid	1510.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1574.000	mm
Gasket Inside Diameter	Gi	1542.000	mm
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	175.77	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	16.000 mm
Basic Gasket Width,	$bo = N/2$	8.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1559.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1560^2 \cdot 150$

= 2869252.500 kgf

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1560 \cdot 2 \cdot 150$$

$$= 209781.328 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(2869252 + 209781 + 0, 0)$$

$$= 3079033.750 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 176 \cdot 7.13 \cdot \pi \cdot 1560 + 0 \cdot 0 \cdot 0$$

$$= 61386.547 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((3079033 + 0 + 4 \cdot 0 / 1560) / 1621, 61387 / 1621)$$

$$= 1899.160 \text{ cm}^2$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1899.160	1956.879	
Radial Distance between Bolts and Edge:	74.000	77.500	
Circ. Spacing between the Bolts:	162.600	162.854	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1957 \cdot 1621 / (176 \cdot \pi (1574 + 1542))$$

$$= 184.388 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 1957$$

$$= 3172611.75 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 3079033.75 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2490 - 1560) / 2$$

$$= 465.1274 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 3079033 \cdot 465 \cdot 1 / (452^2 \cdot 1560) + 0.3 \cdot 1 \cdot 150 \cdot 1560^2 / 452^2$$

$$= 1390.35 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 3172611 \cdot 465 \cdot 1 / (452^2 \cdot 1560)$$

$$= 879.86 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 3079033 / 1957$$

$$= 1573.44 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:



= (Wm2 / Ab)
 = (61387/1957)
 = 31.37 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	1390.35	1407.13	879.86	1407.13
Bolting	1573.44	1621.26	31.37	1621.26

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 0.8203 mm
 Allowed deflection at center of Cover 1.8875 mm

Reqd. Blind Flange Thickness at Center 452.296 mm
 Reqd. Blind Flange Thickness at Gasket 360.418 mm
 Min. Required Flange Thickness per TEMA 342.367 mm

Estimated M.A.W.P. (Operating) 152 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 19376.5 kg
 Estimated Unfinished Weight of Forging at given Thk 19376.5 kg

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.
 This component must be impact tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	1372.08	1407.13	868.29	1407.13
Bolting	1573.44	1621.26	31.37	1621.26

Minimum Req'd Thickness, ambient, no Corrosion: 449.301 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
LHS Blind Flg		150.17	463	3	2650	1407.1	1
LHS Ch Flg		150.05	448	3	1510	1407.1	2
LHS Ch Shell		150.05	92	3	1510	1407.1	3
Main Shell		194.78	118	3	1510	1407.1	4
RHS Ch Shell		150.05	90	3	1510	1407.1	5
RHS Ch Flg		150.05	448	3	1510	1407.1	6
RHS Blind Flg		150.17	463	3	2645	1407.1	7

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind Flg		149.902	152	154	455	452.297	1
LHS Ch Flg		149.902	155	155	440	436.88	2
LHS Ch Shell		149.902	156	162	92	88.2986	3
Main Shell		194.77	199	204	118	115.533	4
RHS Ch Shell		149.902	153	158	90	88.2986	5
RHS Ch Flg		149.902	153	155	440	435.153	6
RHS Blind Flg		149.902	152	154	455	452.296	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Note:

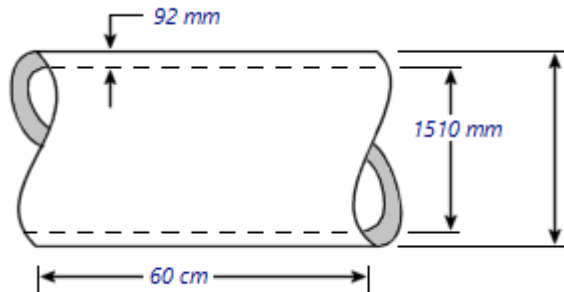
The MAWPs and MAPs for the Exchanger elements all appear to be zero or some could not be calculated. Please fill these values in on the main tubesheet tab.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	150.053 kgf/cm²	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		60.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1510.000 mm	Actual Stress	1278.0 kgf/cm²	
Spec. Min. Thk t _e or s _a		92.0000 mm	Spec. Nominal Thk.	92.000 mm	
Surface Area		31931.1 cm²	SG of Contents	1.000	
Element Volume		1074474.2 cm³	Weight of Contents	1074.0 kgf	
Empty Weight		2153.1 kgf	Operating Weight	3227.1 kgf	
			Comp. All. Ope. Stress	1251.5 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1516 \cdot (\exp(150/(1407 \cdot 1)) - 1) + 3 + 0$
 $= 85.3 + 3 + 0 = 88.3 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 89 + 1516) / 1516) - 0.15$
 $= 156 - 0.15 = 156 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 92 + 1510) / 1510)$
 $= 162 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 150 \cdot 1516 / (1 (1694 - 1516))$
 $= 1277.9807 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$

$$= 150 / (1 \cdot \ln((2 \cdot 89 + 1516) / 1516))$$
$$= 1351.6172 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.743 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

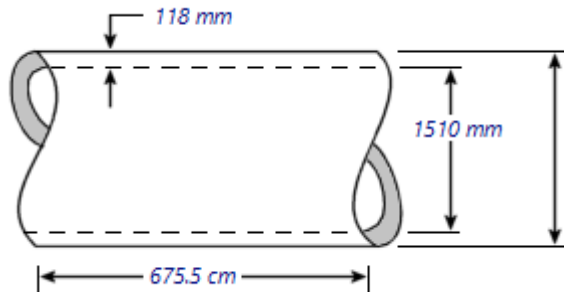
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	194.778 kgf/cm²	Int. Design Temperature	93 °C	
Material		SA-516 70	Ext. Design Temperature	200 °C	
Length		675.500 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1510.000 mm	Actual Stress	1283.8 kgf/cm²	
Spec. Min. Thk t _e , or sa		118.0000 mm	Spec. Nominal Thk.	118.000 mm	
Surface Area		370526.7 cm²	SG of Contents	0.050	
Element Volume		9116371.0 cm³	Weight of Contents	457.4 kgf	
Empty Weight		31596.1 kgf	Operating Weight	32053.6 kgf	
			Comp. All. Ope. Stress	1199.9 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1516 \cdot (\exp(195/(1407 \cdot 1)) - 1) + 3 + 0$$

$$= 113 + 3 + 0 = 116 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1407 \cdot 1 \cdot \ln((2 \cdot 115 + 1516) / 1516) - 0.0076$$

$$= 199 - 0.0076 = 199 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 1407 \cdot 1 \cdot \ln((2 \cdot 118 + 1510) / 1510)$$

$$= 204 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 195 \cdot 1516 / (1 (1746 - 1516))$$

$$= 1283.8409 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 195 / (1 \cdot \ln((2 \cdot 115 + 1516) / 1516))$$
$$= 1378.9381 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 7.248 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

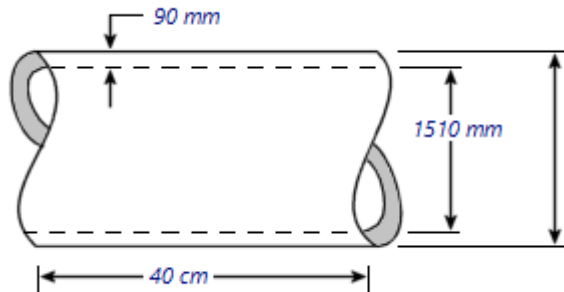
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: RHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	150.053 kgf/cm ²	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		40.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope) S	1407.1	kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1407.1	kgf/cm ²
Inside Diameter	Di	1510.000 mm	Actual Stress	1307.4	kgf/cm ²
Spec. Min. Thk t _e , or sa		90.0000 mm	Spec. Nominal Thk.	90.000	mm
Surface Area		21237.2 cm ²	SG of Contents	1.000	
Element Volume		716316.1 cm ³	Weight of Contents	716.0	kgf
Empty Weight		1402.5 kgf	Operating Weight	2118.5	kgf
			Comp. All. Ope. Stress	1199.9	kgf/cm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1516 \cdot (\exp(150/(1407 \cdot 1)) - 1) + 3 + 0$
 $= 85.3 + 3 + 0 = 88.3 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 87 + 1516) / 1516) - 0.15$
 $= 153 - 0.15 = 153 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 90 + 1510) / 1510)$
 $= 158 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 150 \cdot 1516 / (1 (1690 - 1516))$
 $= 1307.3591 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$

$$= 150 / (1 \cdot \ln((2 \cdot 87 + 1516) / 1516))$$
$$= 1381.0277 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 5.625 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

Hydrotest Pressure Calculation [Shell Side]:

= Test Factor • ShellSide MAWP(St/S)

= 1.25 • 199 (1)

= 248.441 kgf/cm²

Hydrotest Pressure Calculation [Channel Side]:

= Test Factor • Channel Side MAWP(St/S)

= 1.25 • 152 (1)

= 189.642 kgf/cm²

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	1709.6	2386.0	0.716	189.79
Main Shell	1759.9	2386.0	0.738	248.59
RHS Ch Shell	1746.8	2386.0	0.732	189.79

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
T2-Tube outlet	Nozzle	1407.13	1407.13	1.000
T1-Tube Inlet	Nozzle	1407.13	1407.13	1.000
S1-Shell Inlet	Nozzle	1407.13	1407.13	1.000
S2-Shell outlet	Nozzle	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
LHS Blind Flg	1407.13	1407.13	1.000
LHS Ch Flg	1407.13	1407.13	1.000
LHS Ch Shell	1407.13	1407.13	1.000
Main Shell	1407.13	1407.13	1.000
RHS Ch Shell	1407.13	1407.13	1.000
RHS Ch Flg	1407.13	1407.13	1.000
RHS Blind Flg	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	942.17	942.17	1.000
Tubesheet Material	1407.16	1407.16	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
T2-Tube outlet	456.60	2294.24	0.199
T1-Tube Inlet	456.60	2294.24	0.199
S1-Shell Inlet	1406.65	2294.24	0.613
S2-Shell outlet	1406.65	2294.24	0.613



Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		248.44 kgf/cm ²
Shell Thickness	ts	118.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1510.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1784.41 kgf/cm ²
Shell Allowable Stress at Ambient		1784.41 kgf/cm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		189.64 kgf/cm ²
Channel Thickness	tc	92.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1510.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1784.41 kgf/cm ²
Channel Allowable Stress at Ambient		1784.41 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	1548
Tube Wall Thickness	et	3.0480 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	7315.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6755.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Is this a Welded Tube		No
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1413.25 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	30.4500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.4130 mm
Groove Weld Leg	ag	2.4130 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	719.161 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	12174.7 cm ²
Perimeter of Tube Layout	Cp	4316.4 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	21.11 °C
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Tubesheet Material SA-266 2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	1407.16 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1407.16 kgf/cm ²
Thickness of Tubesheet	h	280.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1746.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
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Additional Data for Fixed/Floating Tubesheet Exchangers:

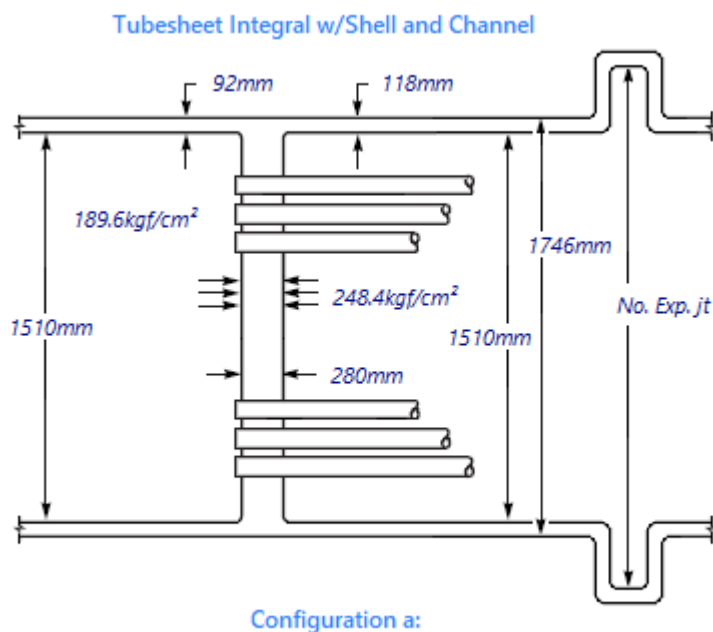
Unsupported Tube Span under consideration	l	937.000 mm
Tube End condition corresponding to Span (l)	k	1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)	Yes
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option	Perform Plastic Calculation	

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
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ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell	- TE-1	Carbon & Low Alloy Steels, Group 1
Shell	- TM-1	Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20670E+07 kgf/cm ²



Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht 21.1 °C 0.0000115191 /°C
 Elastic Mod. at Design Temperature 21.1 °C 0.20670E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len 21.1 °C 0.0000115191 /°C
 Elastic Mod. at Design Temperature 21.1 °C 0.20670E+07 kgf/cm²
 Elastic Mod. at Metal Temp. along Len 21.1 °C 0.20670E+07 kgf/cm²
 Elastic Mod. at Tubsht. Design Temp. 21.1 °C 0.20670E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20670E+07 kgf/cm²
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 13 \cdot (\exp(190/(942 \cdot 1)) - 1) + 0$
 $= 1.44 + 0$
 $= 1.44 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 6755 / \sqrt{(9.52 \cdot 3.05)}$
 $= 1253.676$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 / (19/3.05)$
 $= 0.0880$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot Ey(t / Do)$
 $= 1.6 \cdot 0.088 \cdot 2067025 (3.05/19)$
 $= 46565.957 \text{ kgf/cm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / Fy$
 $= 46566.0/1413.3$
 $= 32.950$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 1413/3304$
 $= 0.428$

Ratio [AE]:

$= Fhe / E$
 $= 46566/2067025$
 $= 0.022528$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.43) + (0.20000\text{E-}04 - 0.002)) / \ln(\ln(1 + 0.20000\text{E-}04) / \ln(1 + 0.002))$$

$$= 0.185$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 1413 (1 + 0.0020) / \ln(1 + 0.0020)^{0.18490803}$$

$$= 4469.191 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1 - 0.43)$$

$$= 0.343$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (3304 \cdot \exp(0.34)) / (0.34^{0.34})$$

$$= 6724.104 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.431.5 - 0.5 \cdot 0.432.5 - 0.433.5$$

$$= 0.309$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	1652	-1.181	0.753E-05	0.375E-05	0.148E-04	-0.319E-05	0.428E+05	0.161E-01
2	826	-4.012	0.355E-06	0.354E-06	0.393E-05	-0.392E-05	0.833E+06	0.215E-01
3	1239	-2.597	0.212E-05	0.206E-05	0.853E-05	-0.816E-05	0.199E+06	0.163E-01
4	1445	-1.889	0.418E-05	0.366E-05	0.115E-04	-0.924E-05	0.949E+05	0.729E-02
5	1548	-1.535	0.567E-05	0.422E-05	0.131E-04	-0.787E-05	0.642E+05	0.159E-02
6	1497	-1.712	0.488E-05	0.400E-05	0.123E-04	-0.884E-05	0.782E+05	0.339E-02
7	1523	-1.623	0.526E-05	0.413E-05	0.127E-04	-0.844E-05	0.709E+05	0.105E-02
8	1536	-1.579	0.546E-05	0.418E-05	0.129E-04	-0.818E-05	0.675E+05	0.227E-03
9	1529	-1.601	0.536E-05	0.416E-05	0.128E-04	-0.832E-05	0.692E+05	0.423E-03
10	1532	-1.590	0.541E-05	0.417E-05	0.128E-04	-0.825E-05	0.683E+05	0.101E-03
11	1534	-1.585	0.543E-05	0.418E-05	0.128E-04	-0.821E-05	0.679E+05	0.625E-04
12	1533	-1.587	0.542E-05	0.418E-05	0.128E-04	-0.823E-05	0.681E+05	0.193E-04
13	1534	-1.586	0.543E-05	0.418E-05	0.128E-04	-0.822E-05	0.680E+05	0.215E-04
14	1533	-1.587	0.543E-05	0.418E-05	0.128E-04	-0.823E-05	0.681E+05	0.110E-05
15	1533	-1.587	0.542E-05	0.418E-05	0.128E-04	-0.823E-05	0.681E+05	0.911E-05
16	1533	-1.587	0.542E-05	0.418E-05	0.128E-04	-0.823E-05	0.681E+05	0.400E-05

Factor of Safety [FS]:

$$\text{Since } F_{ic}[1534] \geq F_y[1413], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 1534/1.67$$

$$= 920.081 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t/Do)$$

$$= 2 \cdot 920(3.05/19)$$

$$= 294.426 \text{ kgf/cm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.6743 mm
Allowable Internal Pressure at Corroded thickness	363.36 kgf/cm ²
Required Internal Design Pressure	189.64 kgf/cm ²
Allowable External Pressure at Corroded thickness	294.43 kgf/cm ²
Required External Design Pressure	248.44 kgf/cm ²
Required Thickness due to Shell Side pressure	2.6743 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:
 $= l_{tx} / h$ (modified for corrosion if present)
 $= 0/280$
 $= 0.0000$ (must be $0 \leq p \leq 1$)

Effective tube hole diameter [d*]:
 $= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (p); dt - 2 \cdot tt)$
 $= \max(19 - 2 \cdot 3.05 (2067025/2067025) \cdot (942/1407) \cdot (0); 19 - 2 \cdot 3.05)$
 $= 19.0500 \text{ mm}$

Equivalent outer tube limit circle diameter [Do]:
 $= 2 \cdot ro + dt$
 $= 2 \cdot 719 + 19$
 $= 1457.372 \text{ mm}$

Determine the basic ligament efficiency for shear [μ]:
 $= (p - dt) / p$
 $= (30.5 - 19) / 30.5$
 $= 0.374$

Equivalent outer tube limit radius [ao]:
 $= Do / 2$
 $= 1457/2$
 $= 728.6860 \text{ mm}$

Effective tube pitch [p*]:
 $= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$
 $= 30.5 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1457 \cdot 30.5) / (\pi \cdot 1457^2)) }$
 $= 30.4500 \text{ mm}$

Effective ligament efficiency for bending [μ^*]:
 $= (p^* - d^*) / p^*$
 $= (30.5 - 19) / 30.5$
 $= 0.3744$

Ratio [ps]:
 $= as / ao$
 $= 755/729$
 $= 1.0361$

Ratio [pc]:
 $= ac / ao$
 $= 755/729$
 $= 1.0361$

Compute parameter [xt]:
 $= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$
 $= 1 - 1548 ((19 - 2 \cdot 3.05) / (2 \cdot 729))^2$
 $= 0.8777$

Determine parameter [xs]:
 $= 1 - Nt(dt / (2 \cdot ao))^2$
 $= 1 - 1548 (19 / (2 \cdot 729))^2$
 $= 0.7355$

Determine the value [h'g]:
 $= \max((hg - CATC); 0)$ (For pressure only cases)
 $= \max((0 - 0); 0)$
 $= 0.000 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts (Ds + ts) Es / L$$

$$= \pi \cdot 118 (1510 + 118) 2067025 / 6755$$

$$= 1846742.2500 \text{ kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (Dt - tt) Et / L$$

$$= \pi \cdot 3.05 (19 - 3.05) 2067025 / 6755$$

$$= 468.8771 \text{ kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 1846742 / (1548 \cdot 469)$$

$$= 2.54435$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kj)$$

$$= 1 / (1 + 1846742/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1510 + 118)118)$$

$$= 0.0041 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.0041 \cdot 2067025 \cdot 118^3 / (6(1 - 0.3^2))$$

$$= 2579804160.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1510 + 118) \cdot 2.58E+09 / 280^3 \cdot (1 + 280 \cdot 0.0041 + (280 \cdot 0.0041)^2 / 2)$$

$$= 3255108.5000 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 1510^2 / (4 \cdot 2067025 \cdot 118) \cdot (1 - 1510 \cdot 0.3 / (2(1510 + 118)))$$

$$= 0.20119E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((Dc + tc)tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1510 + 92)92)$$

$$= 0.0047 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot Ec \cdot tc^3 / (6(1 - \nu_c^2))$$

$$= 0.0047 \cdot 2067025 \cdot 92^3 / (6(1 - 0.3^2))$$

$$= 1395877120.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1510 + 92 / \cos(0)) \cdot 1395877120 / 280^3 \cdot (1 + 280 \cdot 0.0047 \cdot (1 + 1) / 2 + (280 \cdot 0.0047)^2 / 2 \cdot 1)$$

$$= 1958757.0 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot Ec \cdot Tc) \cdot (1 - Dc \cdot \nu_c / (2(Dc + tc)))$$

$$= 1510^2 / (4 \cdot 2067025 \cdot 92) \cdot (1 - 1510 \cdot 0.3 / (2(1510 + 92)))$$

$$= 0.25737E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 9.195403 ; \mu^* = 0.374384 ; E^* = 784439 \text{ kgf/cm}^2$$

$$E^*/E = 0.379501 ; \nu^* = 0.317620$$

Note: As h/p (9.195) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:

$$= \left(\left(\frac{24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2}{E^* \cdot L \cdot H^3} \right) \right)^{1/4}$$

$$= \left(\left(\frac{24(1 - 0.32^2)1548 \cdot 2067025 \cdot 3.05}{(19 - 3.05)729^2(784439 \cdot 6755 \cdot 280^3)} \right) \right)^{1/4}$$

$$= 1.9801$$

Values from Table 4.18.3

$$Z_d = 0.179335 ; Z_v = 0.163904 ; Z_m = 0.681118$$

$$Z_a = 0.785445E+00 ; Z_w = 0.163904$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio $[K]$:

$$= A / D_o$$

$$= 1746/1457$$

$$= 1.1980$$

Compute coefficient $[F]$:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 784439 \cdot (3255108 + 1958757 + 2067025 \cdot \ln(1.2))$$

$$= 4.8604$$

Compute parameter $[\phi]$:

$$= (1 + \nu^*)F$$

$$= (1 + 0.32)4.86$$

$$= 6.4042$$

Compute parameter $[Q_1]$:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.04 - 1 - 6.4 \cdot 0.16) / (1 + 6.4 \cdot 0.68)$$

$$= -0.189027$$

Compute Parameter $[Q_{z1}]$:

$$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.18 + -0.19 \cdot 0.16) / 2 \cdot 1.98^4$$

$$= 1.1403$$

Compute parameter $[Q_{z2}]$:

$$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.16 + -0.19 \cdot 0.68) / 2 \cdot 1.98^4$$

$$= 0.2702$$

Compute parameter $[U]$:

$$= (Z_w + (\rho_s - 1)Z_m)X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.16 + (1.04 - 1)0.68)1.98^4 / (1 + 6.4 \cdot 0.68)$$

$$= 0.5404$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor $[y_b]$:

$$= 0 \text{ (config a)}$$

Compute Parameter $[y]$:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter $[\omega_s]$:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.04 \cdot 2.58E+09 \cdot 0.0041 \cdot 0.002011892 (1 + 280 \cdot 0.0041)$$

$$= 48205.9844 \text{ mm}^2$$

Calculate parameter $[\omega^*s]$:



$$\begin{aligned}
 &= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega s \\
 &= 729^2(1.04^2 - 1) \cdot (1.04 - 1) / 4 - 48206 \\
 &= -47853.5234 \text{ mm}^2
 \end{aligned}$$

Parameter [wc]:

$$\begin{aligned}
 &= \rho c \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\
 &\quad \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)) \\
 &= 1.04 \cdot 1395877120 [0.002573702 (1 + 0.0047 \cdot 1 \cdot 280) - \\
 &\quad 0.000 (1/280 + 1 \cdot 0.0047/2)) \\
 &= 40994.0469 \text{ mm}^2
 \end{aligned}$$

Parameter [ω*c]:

$$\begin{aligned}
 &= ao^2 [(\rho c^2 + 1) \cdot (\rho c - 1) / 4 - (ps - 1) / 2] - \omega c \\
 &= 729^2 [(1.04^2 + 1) \cdot (1.04 - 1) / 4 - (1.04 - 1) / 2] - 40994 \\
 &= -40641.5859 \text{ mm}^2
 \end{aligned}$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$\begin{aligned}
 &= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - \\
 &\quad [(ps^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(Dj^2 - (Ds)^2) / Do^2] \} \\
 &= 248 \{ 0.74 + 2(1 - 0.74) 0.3 + [2/2.54 \cdot \\
 &\quad (1510/1457)^2] 0.3 - [(1.04^2 - 1) / (1 \cdot 2.54)] - \\
 &\quad [(1 - 1) / (2 \cdot 1 \cdot 2.54)] [(0^2 - (1510)^2) / 1457^2] \} \\
 &= 277.8706 \text{ kgf/cm}^2
 \end{aligned}$$

Pressure [P't]:

$$\begin{aligned}
 &= [xt + 2(1 - xt) vt + 1 / (J \cdot Kst)] Pt \\
 &= [0.88 + 2(1 - 0.88) 0.3 + 1 / (1 \cdot 2.54)] 190 \\
 &= 254.8990 \text{ kgf/cm}^2
 \end{aligned}$$

Pressure [Pγ]:

$$\begin{aligned}
 &= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\
 &= 1548 \cdot 469 \cdot 0 / (3.14 \cdot 729^2) \\
 &= 0.0000 \text{ kgf/cm}^2
 \end{aligned}$$

Pressure [Pw]:

$$\begin{aligned}
 &= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2) \\
 &= -0 \cdot 0.54 \cdot 0 / (2 \cdot \pi \cdot 729^2) \\
 &= -0.0000 \text{ kgf/cm}^2
 \end{aligned}$$

Calculate the pressure [Prim]:

$$\begin{aligned}
 &= - (U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt) \\
 &= - (0.54/729^2) \cdot (-47854 \cdot 248 - -40642 \cdot 190) \\
 &= 4.2558 \text{ kgf/cm}^2
 \end{aligned}$$

Calculate the pressure [Pw]:

$$\begin{aligned}
 &= (U/ao^2) \cdot (\omega s \cdot P*s - \omega c \cdot P*c) \\
 &= (0.54/729^2) \cdot (48206 \cdot 0 - 40994 \cdot 0) \\
 &= 0.0000 \text{ kgf/cm}^2
 \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned}
 &= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot \\
 &\quad (P's - P't + P\gamma + Pw + Pw + Prim) \\
 &= 0.1000E+01 \cdot 2.54 / (1 + 1 \cdot 2.54 (1.14 + (1.04 - \\
 &\quad 1) 0.27)) \cdot (278 - 255 + 0 + 0 + -0 + 4.26) \\
 &= 17.6448 \text{ kgf/cm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned}
 &= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega s \cdot P*s - \omega c \cdot P*c)) + \\
 &\quad W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm) \\
 &= [((-47854 \cdot 248 - -40642 \cdot 190) -
 \end{aligned}$$

$$\begin{aligned} & (48206 \cdot 0 - 40994 \cdot 0) \cdot 0.01 + \\ & 0 \cdot 0 / (2 \cdot \pi) / (1 + 6.4 \cdot 0.68) \\ & = -7798.2075 \text{ kgf} \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned} & = Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\ & = -0.19 + 2 \cdot -7798 / (17.6 \cdot 729^2) \\ & = -0.35549 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.177747

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} & = (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\ & = (1.5 \cdot 0.18 / 0.37) \cdot (2 \cdot 729 / (280 - 0))^2 \cdot 17.6 \\ & = 340.4209 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 2549 = 3633 \text{ kgf/cm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ]:

$$\begin{aligned} & = (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ & = (1.5 \cdot 0.018 / 0.37) \cdot (2 \cdot 729 / (11.3 - 0))^2 \cdot 3.09 \\ & = 3632.5200 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} & = h + Cats + Catc \\ & = 11.3 + 0 + 0 \\ & = 11.2695 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [Tlimit]:

$$\begin{aligned} & = (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy) \\ & = (2 \cdot 0.37 \cdot 280 / 729) \cdot 1937 \\ & = 557.4413 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Tau limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned} \text{Tubesheet thickness (Incl. Corr.)} &= 3.8100 \text{ mm} \\ \text{Tubesheet Shear Stress} &= 282.1483 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$\begin{aligned} & = \max(HreqB; HreqS) \\ & = \max(11.3; 3.81) \\ & = 11.2695 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 F_{tmin} = 0.93, F_{tmax} = 1.02

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} & = ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs) \\ & = ((248 \cdot 0.74 - 190 \cdot 0.88) - (17.6) \cdot 0.93) / \\ & \quad (0.88 - 0.74) \\ & = -0.9781 \text{ kgf/cm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} & = ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs) \\ & = ((248 \cdot 0.74 - 190 \cdot 0.88) - (17.6) \cdot 1.02) / \\ & \quad (0.88 - 0.74) \\ & = -12.5213 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 12.5 \text{ kgf/cm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta T \cdot Sy, t = 1342.5896 \text{ kgf/cm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.18 + -0.36 \cdot 0.16)1.98^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (\frac{dt^2}{4} + (dt - 2 \cdot tt)^2) }) / 4$$

$$= (\sqrt{ (\frac{19^2}{4} + (19 - 2 \cdot 3.05)^2) }) / 4$$

$$= 5.7593 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Et / S_{yt})}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 2067025 / 1413)}$$

$$= 157.1179$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 937 / 5.76$$

$$= 162.6940$$

Buckling Allowable Stress [Stb]:

$$= (1 / F_s) \cdot (\pi^2 \cdot Et / F_t^2) (C_t \leq F_t)$$

$$= (1 / 2) \cdot (\pi^2 \cdot 2067025 / 163^2)$$

$$= 1342.590 \text{ kgf/cm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 12.5 \cdot 1.53$$

$$= 19.19 \text{ kgf}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 3.05) 3.05$$

$$= 1.532 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (19 - 3.05) 1343$$

$$= 2057.225 \text{ kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.41 (19 + 0.67 \cdot 2.41) 1343$$

$$= 1787.8877 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 1343 / 1343$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.55 \cdot \pi \cdot 2.41 (19 + 0.67 \cdot 2.41) 1343$$

$$= 1156.8684 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2 (\sqrt{ (0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d } - 0.75 \cdot d_o)$$

$$= 2 (\sqrt{ (0.75 \cdot 19)^2 + 1.07 \cdot 3.05 \cdot (19 - 3.05) 1 \cdot 1 } - 0.75 \cdot 19)$$

$$= 3.445 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm
 Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 2057 is >= tube strength 2057 kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot S_{y,s} = 0.95 \cdot 2651 = 3778 \text{ kgf/cm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 729^2 / ((1510 + 118) 118) \cdot [17.6 + (1.04^2 - 1)(248 - 190)] + 755^2 \cdot 190 / ((1510 + 118) 118)$
 $= 623.4398 \text{ kgf/cm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 118 mm, the minimum Shell length adjacent to the tubesheet should be 759.805 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	340.4 <=	3632.7 kgf/cm ²	Pass
Tubesheet Shear Stress	47.5 <=	1937.5 kgf/cm ²	Pass
Maximum Tube Stress	12.5 <=	1342.6 kgf/cm ²	Pass
Minimum Tube Stress (Buckling)	-12.5 <=	-1342.6 kgf/cm ²	Pass
Maximum Force on any one Tube	19.2 <=	2057.2 kgf	Pass
Axial Membrane Stress in Shell	623.4 <=	3777.8 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	11.270	280.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	2.413	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	2.413	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	198.787	...	2041	3633	458	1937	Ps+Pt-Th	D1 Ok
D2uc	216.968	...	2381	3633	506	1937	Ps+Pt-Th	D2 Ok
D3uc	11.270	...	340	3633	47	1937	Ps+Pt-Th	D3 Ok
Max:	216.9680	... mm	0.655	0.261	(Str. Ratio)			

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	Ten	Allwd	Cmp	Allwd	Ten	Allwd	Cmp	Allwd	Pass	Fail
D1uc	54	3777	...	...	...	...	...	...	Ok	
D2uc	570	3777	...	...	...	...	...	...	Ok	
D3uc	623	3777	...	...	...	...	...	...	Ok	
Max RATIO	0.165			...		...		...		

Tube, Shell and Channel Stress Summary:

Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Shell Stress	Shell Allwd	Channel Stress	Channel Allwd	Pass	Fail
D1uc	179	1342	-179	-1342	275	2057	...	3777	...	3777	Ok	
D2uc	171	1342	-148	-1342	262	2057	...	3777	...	3777	Ok	
D3uc	13	1342	-13	-1342	19	2057	...	3777	...	3777	Ok	
Max RATIO	0.134		0.134		0.134		...		...			

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	216.968	280.000	Ok
Tube thickness :	2.674	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	2.413	Ok
Tube-tubesheet groove weld leg :	1.829	2.413	Ok

Min Shell length of thk, (118.000) adj. to tubesheet: 759.805 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.749 mm
 Due to Pressure + Thermal Shell Compresses by : -0.749 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure UnCorroded	Due to Pressure Corroded	Due to Thermal Exp. UnCorroded	Due to Thermal Exp. Corroded	Computed UnCorroded	Delta Ms Corroded
1	0.0450	...	...	...	...	...
2	-0.7491	...	...	...	...	...
3	-0.7041	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	452	No Calc	No Calc
20	30	No Calc	...	437	No Calc	No Calc
30	Tsht	121.997	1694	89	No Calc	No Calc
Tsht	40	No Calc	...	89	No Calc	No Calc
40	Tsht	675.501	1746	115	No Calc	No Calc
Tsht	50	No Calc	...	115	No Calc	No Calc
50	60	102.002	1690	87	No Calc	No Calc
60	70	No Calc	...	437	No Calc	No Calc
70	80	No Calc	...	452	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	455	360.418	...	No Calc
20	30	440	436.88	...	No Calc
30	Tsht	92	No Calc	...	160
Tsht	40	92	No Calc	...	No Calc
40	Tsht	118	No Calc	...	137
Tsht	50	118	No Calc	...	No Calc
50	60	90	No Calc	...	160
60	70	440	435.153	...	No Calc
70	80	455	360.418	...	No Calc

Minimum

137

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	121.997	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	675.501	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	102.002	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	19791.8	...	19791.8	...	...
20	30	13668.1	1131004	13598.7	1133571	...
30	40	2153.15	1074474	2086.84	1083030	...
40	50	31596.1	9116371	30849.6	9212697	...
50	60	1402.47	716316	1358.27	722020	...
60	70	13660.5	1131004	13591.1	1133571	...
70	80	19717.2	...	19717.2	...	...
Total		101989	13169170.00	100993	13284889.00	0



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Liqd	...	22.75	...	...	Liquid: 10
20	Liqd	1130.49	31	...	...	Liquid: 20
30	Liqd	1073.99	30	...	...	Liquid: 30
30	Nozl	495.241	30	88.2	...	T2-Tube outlet
30	Nozl	495.241	30	-88.2	...	T1-Tube Inlet
30	Wght	1000	...	...	...	Datasheet weigh
40	Sadl	754.076	75.5	95.65	...	Left Saddle
40	Sadl	754.076	600	95.65	...	New Saddle
40	Liqd	457.438	337.75	...	...	Liquid: 40
40	Insl	752.163	337.75	...	...	Hot Insulation
40	Nozl	2451.78	150	100.743	...	S1-Shell Inlet
40	Nozl	2577.31	500	-100.743	...	S2-Shell outlet
40	Wght	9999.82	350	...	...	margin
40	Wght	34999.4	350	...	...	datasheet weigh
50	Liqd	715.993	20	...	...	Liquid: 50
60	Liqd	1130.49	31	...	...	Liquid: 60
70	Liqd	...	22.75	...	...	Liquid: 70
30	FTsh	4238.17	74	...	...	
30	Tube	13447.7	425.75	...	...	Tubes
30	RTsh	4238.17	777.5	...	...	Tubesheet
40	Bafl	125.278	109.6	...	...	Baffles
40	Bafl	125.278	203.4	...	...	Baffles
40	Bafl	125.278	297.2	...	...	Baffles
40	Bafl	125.278	391	...	...	Baffles
40	Bafl	125.278	484.8	...	...	Baffles
40	Bafl	125.278	578.6	...	...	Baffles
40	Bafl	125.702	672.4	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	1508.2
Liquid	4508.4
Insulation	752.2
Nozzles	6019.6
Weights	45999.2
Exchanger Components	22801.4
Liquid in Tubes	1491.7
Sum of the Detail Weights	83080.6 kg

Weight Summation Results: (kg)



	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	101989.2	101989.2	101989.2	101989.2	101989.2	101989.2
Saddles	1508.2	1508.2	1508.2	1508.2	1508.2	1508.2
Nozzles	6019.6	6019.6	6019.6	6019.6	6019.6	6019.6
Wld Weights	45999.2	45999.2	45999.2	45999.2	45999.2	45999.2
Exchanger	22801.4	22801.4	22801.4	22801.4	22801.4	22801.4
Insulation	...	...	752.2	752.2	752.2	752.2
Ope. Liquid	...	...	...	...	...	4508.4
Tube Ope Lqd	...	...	...	...	...	1491.7
Test Liquid	...	13163.2	...	...	...	...
Tube Tst Lqd	...	1491.7	...	...	...	...
Totals	178317.5	192972.5	179069.7	179069.7	179069.7	185069.8

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	178317.5 kg
Shop Test Wt.	- Fabricated weight + water (full)	192972.5 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	179069.7 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	179069.7 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	179069.7 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	185069.8 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	184074.0 kg
Field Test Wt.	- Empty weight + test medium (full)	179565.6 kg

Exchanger Tube Data

Volume of Exchanger tubes :	1492395.4 cm ³
Weight of Ope Liq in tubes :	1491.7 kg
Weight of Water in tubes :	1491.7 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	93700.4	...
20	30	84331.1	29791.1
30	40	31931.1	28462.8
40	50	370527	320444
50	60	21237.2	18975.2
60	70	84331.1	29791.1
70	80	93419.7	...

Total	779477.2	427464.3 cm ²
Total	839.0	460.1 ft ²

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
T2-Tube outlet	242.82	260.14	85	1500	GR 1.1	...	...	...	...
T1-Tube Inlet	242.82	260.14	85	1500	GR 1.1	...	...	...	...
S1-Shell Inlet	240.20	260.14	93	1500	GR 1.1	...	...	...	...
S2-Shell outle	240.20	260.14	93	1500	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 240.204 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 260.136 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 242.824 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 260.136 kgf/cm²

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 cm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.92)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.579
Structure Height to Diameter ratio	5.735

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5 : Z_g = 58979 \text{ cm}$

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 130 + 0 = 130 \text{ cm}$
 $= 4.27 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because $z (4.27 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15/1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: [Kzt]:
 $K_1 = 0, K_2 = 0, K_3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf } [162 \text{ kgf/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.58 \cdot 15.6$$

$$= 254 \text{ lbs. } [115 \text{ kgf}]$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
LHS Blind Flg	130.0	0.000	0.000	0.000	1.035	1.000	161.847
LHS Ch Flg	130.0	0.000	0.000	0.000	1.035	1.000	161.847
LHS Ch Shell	130.0	0.000	0.000	0.000	1.035	1.000	161.847
Main Shell	130.0	0.000	0.000	0.000	1.035	1.000	161.847
RHS Ch Shell	130.0	0.000	0.000	0.000	1.035	1.000	161.847
RHS Ch Flg	130.0	0.000	0.000	0.000	1.035	1.000	161.847
RHS Blind Flg	130.0	0.000	0.000	0.000	1.035	1.000	161.847

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Datasheet wei	0.00	130.00	161.85	0.00
margin	0.00	130.00	161.85	0.00
datasheet wei	0.00	130.00	161.85	0.00

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	130	318	14469	161.847	126.757
20	30	130	181.2	11234.4	161.847	98.4197
30	40	130	203.28	12196.8	161.847	106.851
40	50	130	233.52	157743	161.847	1381.92
50	60	130	202.8	8112	161.847	71.0657
60	70	130	181.2	11234.4	161.847	98.4197
70	80	130	317.4	14441.7	161.847	126.517

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(8.610.75)$
 = 0.101 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.1$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.22 \cdot 182566 \cdot 1.5$
 = 17854.939 kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.22 \cdot 182566$
 = 7935.5 kgf

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	132.5	20285.1	2982.39
20	30	75.5	20285.1	1699.4
30	40	75.5	20285.1	1699.4
40	Sad1	75.5	20285.1	1699.4
Sad1	50	75.5	20285.1	1699.4
40	50	75.5	20285.1	1699.4



50	60	75.5	20285.1	1699.4
60	70	75.5	20285.1	1699.4
70	80	132.25	20285.1	2976.76

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	533.568 cm
Center of Gravity of the Liquid	495.619 cm
Center of Gravity of the Insulation	533.567 cm
Center of Gravity of the Nozzles	461.447 cm
Center of Gravity of the Added Weights (Operating)	536.296 cm
Center of Gravity of the Added Weights (Empty)	536.296 cm
Center of Gravity of the Tubesheet(s)	533.568 cm
Center of Gravity of the Tubes	533.568 cm
Center of Gravity of the Baffles	586.954 cm
Center of Gravity of Bare Shell New and Cold	523.668 cm
Center of Gravity of Bare Shell Corroded	523.675 cm
Vessel CG in the Operating Condition	525.788 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	526.467 cm
Vessel CG in the Test Condition	526.121 cm

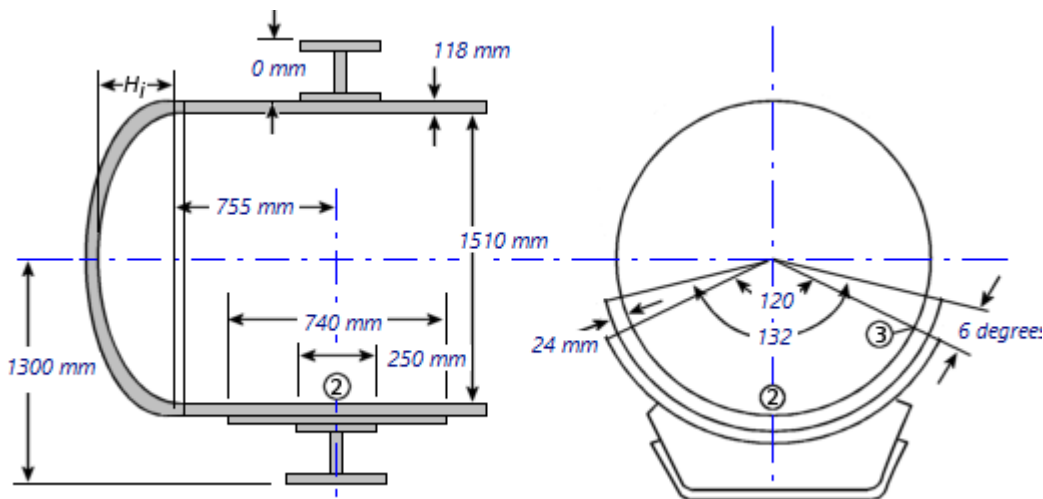
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	814.00	mm
Shell Thickness used in this Case	t	118.000	mm
Stiffened Vessel Length per 4.15.6	L	675.50	cm
Distance from Saddle to Vessel tangent	a1 or a	755.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	740.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		28699.50	kgf
Tube Bundle Pull out Force Impact Factor		1.700	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force	91184.73	kgf
Pressure used in Analysis	P	0.000 kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-36.48	-2518.57
Long. Stress at Top of Midspan	36.48	1407.13
Long. Stress at Bottom of Midspan	36.48	1407.13
Long. Stress at Top of Saddles	12.31	1407.13
Long. Stress at Bottom of Saddles	-6.82	-2518.57
Long. Stress at Bottom of Saddles	6.82	1407.13
Tangential Shear in Shell	86.29	1125.70
Circ. Stress at Horn of Saddle	53.84	2110.69

Circ. Compressive Stress in Shell	6.66	1407.13
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Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 28700 • 1.7 • 1300/5245
 = 12092.6 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 79092 + max(12093, 0, 0, 0)
 = 91184.7 kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	91938.80	kgf
Transverse Shear Load	0.00	kgf
Longitudinal Shear Load	48789.15	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 1500.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0471	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0387			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 = $-91185 \cdot 75.5 [1 - (1 - 75.5/676 + (81.4^2 - 0^2)/(2 \cdot 75.5 \cdot 676))/(1 + (4 \cdot 0)/(3 \cdot 676))]$
 = -3222.5 kgf-m.

Moment per Equation 4.15.2 [M2]:
 = $Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 = $91185 \cdot 676/4 ((1 + 2(81.4^2 - 0^2)/(676^2))/(1 + (4 \cdot 0)/(3 \cdot 676)) - 4 \cdot 75.5/676)$
 = 89616.0 kgf-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 = $0 \cdot 814/(2 \cdot 118) - 89616/(\pi \cdot 814^2 \cdot 118)$
 = -36.48 kgf/cm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 = $0 \cdot 814/(2 \cdot 118) + 89616/(\pi \cdot 814^2 \cdot 118)$
 = 36.48 kgf/cm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 = $P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$
 = $0 \cdot 814/(2 \cdot 118) - 3223/(0.11 \cdot \pi \cdot 814^2 \cdot 118)$
 = 12.31 kgf/cm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:
 = $P \cdot Rm/(2t) + M1/(K1* \cdot \pi \cdot Rm^2 \cdot t)$
 = $0 \cdot 814/(2 \cdot 118) + 3223/(0.19 \cdot \pi \cdot 814^2 \cdot 118)$
 = -6.82 kgf/cm²

Maximum Shear Force in the Saddle (4.15.3) [T]:
 = $Q(L - 2a)/(L + (4 \cdot hm/3))$

$$= 91185 (676 - 2 \cdot 75.5) / (676 + (4 \cdot 0/3))$$

$$= 70801.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 70801 / (814 \cdot 118)$$

$$= 86.29 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(814 \cdot 118)}$$

$$= 241.740 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250 + 1.56 \cdot \sqrt{(814 \cdot 118)}, 2 \cdot 75.5)$$

$$= 733.48 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 91185 \cdot 0.1 / (733(118 + 1 \cdot 24))$$

$$= -6.66 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -91185 / (4(118 + 1 \cdot 24)733) -$$

$$3 \cdot 0.047 \cdot 91185 / (2(118 + 1 \cdot 24)^2)$$

$$= -53.84 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1500.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(755 \cdot 118)} + 740)118^3) / 12$$

$$= 16507.363 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 16507 + 1423(107 - 59)^2$$

$$= 49772.742 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1205.6	118.0	59.0	1422.6	8393582.0	0.498E+05
Wearplate	740.0	24.0	130.0	177.6	2308800.2	996.
Web	24.0	363.0	323.5	87.1	2818332.5	0.503E+05
BasePlate	275.0	40.0	525.0	110.0	5775000.5	0.192E+06
Totals	...	...	...	1797.4	19295714.0	0.293E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 19295714 / 1797$$

$$= 107.356 \text{ mm}$$

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120/2
 = 120.0

Saddle Splitting Coefficient [K1]:
 = (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
 = (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))
 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.2 • 91185
 = 18558.0703 kgf

Tension Stress, St = (Fh/As) = 49.5253 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1590.6736 kgf/cm²

Saddle Splitting Dimension [d]:
 = B - R • sin(theta/2)/(theta/2 in radians)
 = 1300 - 755 • sin(120/2)/1.05
 = 675.620 mm

Bending Moment, M = Fh * d = 12538.2197 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 45.9321 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1767.4150 kgf/cm²

Minimum Thickness of Baseplate per Moss:

= $\sqrt{\frac{3(Q + \text{SaddleWt})\text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$
 = $\sqrt{\frac{3(91185 + 754)275}{4 \cdot 1500 \cdot 1767}}$
 = 26.744 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120/2)(755 + 118 + 24)
 = 1553.650 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 1554/(5 - 1)
 = 388.412 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 388 • 275/2
 = 534.067 cm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 91939/(1500 • 275)
 = 22.288 kgf/cm²

Axial Load [P]:
 = Ap • Bp
 = 534 • 22.3
 = 11903.390 kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 54.2 + 46.6
 = 100.849 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 11903/101
 = 118.031 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	250.0	...	60.0	...	0.312E+04

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 250 - 24 \\
 &= 226.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0/101 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 250/2 \\
 &= 125.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(3125/101)} \\
 &= 55.666 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1300 - \cos(120/2)(755 + 118 + 24) - 40 \\
 &= 811.500 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120/2)(755 + 118 + 24) \\
 &= 1553.650 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 1500 - 50.8 \\
 &= 1449.200 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (1554 - 1449) / 2 \\
 &= 52.225 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{Lv^2 + Dd^2} \\
 &= \sqrt{812^2 + 52.2^2} \\
 &= 813.179 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2651)} \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 813 / 55.7 \\
 &= 14.608
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 48789 / (2 \cdot 1500) \cdot 388 \cdot 813 / 2 \\
 &= 2568.334 \text{ kgf-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($14.6 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.6)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (14.6) / (8 \cdot 123) - (14.6^3) / (8 \cdot 123^3))$$

$$= 1539 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_a + (R_m \cdot C_1 / I) / S_b$$

$$= 118 / 1539 + (2568 \cdot 125 / 31250000) / 1767$$

$$= 0.658$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	226.0	0.0	54.2	0.0	0.312E+04
Web	388.4	24.0	0.0	93.2	0.0	44.7
Totals	...	...	...	147.5	...	0.317E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 147$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250 / 2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1300 - \cos(897/2) (40 + 0 + 0) - 0$$

$$= 363.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3167 / 147)}$$

$$= 46.343 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 363 / 46.3$$

$$= 7.833$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 48789 / (2 \cdot 1500)$$

$$= 16.263 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 16.3 \cdot 388 \cdot 363$$

$$= 2292.990 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2293 / 253358$$

$$= 905.038 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.83 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.83)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (7.83) / (8 \cdot 123) - (7.83^3) / (8 \cdot 123^3))$$

$$= 1565 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_a + (M_{base,c} \cdot C_1 / I) / S_b$$

$$= 162 / 1565 + (2293 \cdot 125 / 3167) / 1767$$

$$= 0.615$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-193 B7	
Bolt Allowable Stress	Stba	1753.92 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BlArea	14.0743 cm ²
Saddle Load QO (Weight)	QO	91938.8 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	48789.1 kgf
Maximum Transverse Force	Ft	0.0 kgf
Maximum Longitudinal Force	Fl	48789.1 kgf
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 48789 / (14.1 \cdot 4)$$

$$= 866.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (14.1 \cdot 4)$$

$$= 0.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 48789 / (1052 • 4)
 = 11.5905 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1300 • 0 + 0
 = 0.00 kgf-m.

Eccentricity (e):
 = Rmom / QO
 = 0/91939
 = 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 11.6, 0]
 = 11.5905 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

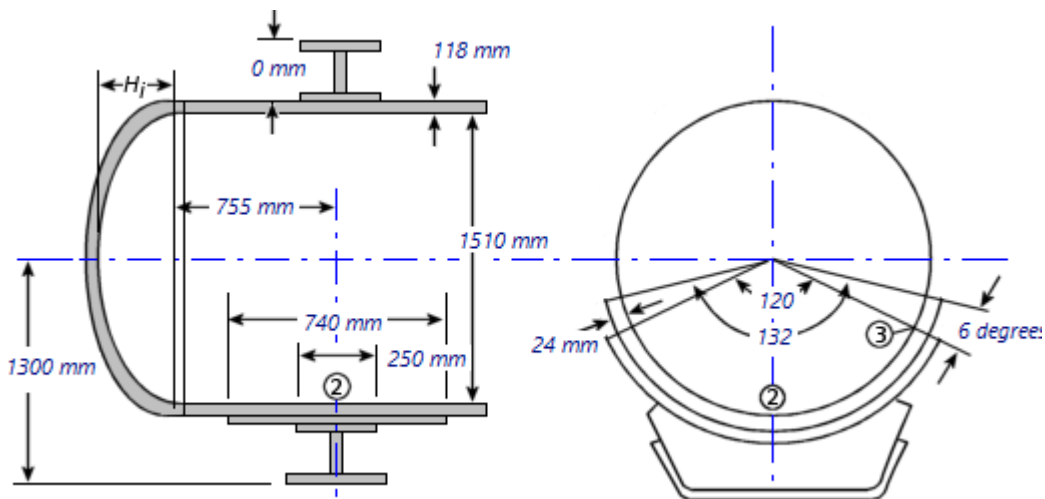
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	815.50	mm
Shell Thickness used in this Case	t	118.000	mm
Stiffened Vessel Length per 4.15.6	L	675.50	cm
Distance from Saddle to Vessel tangent	a1 or a	755.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	740.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		28699.50	kgf
Tube Bundle Pull out Force Impact Factor		1.700	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force	90686.79	kgf
Pressure used in Analysis	P	0.000 kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-37.10	-2518.57
Long. Stress at Top of Midspan	37.10	1407.13
Long. Stress at Bottom of Midspan	37.10	1407.13
Long. Stress at Top of Saddles	12.45	1407.13
Long. Stress at Bottom of Saddles	-6.90	-2518.57
Long. Stress at Bottom of Saddles	6.90	1407.13
Tangential Shear in Shell	87.90	1125.70
Circ. Stress at Horn of Saddle	55.48	2110.69

Circ. Compressive Stress in Shell	6.82	1407.13
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Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 28700 • 1.7 • 1300/5245
 = 12092.6 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 78594 + max(12093, 0, 0, 0)
 = 90686.8 kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	91440.87	kgf
Transverse Shear Load	0.00	kgf
Longitudinal Shear Load	48789.15	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 1500.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0470	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0386			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 = $-90687 \cdot 75.5 [1 - (1 - 75.5/676 + (81.6^2 - 0^2)/(2 \cdot 75.5 \cdot 676))/(1 + (4 \cdot 0)/(3 \cdot 676))]$
 = -3188.5 kgf-m.

Moment per Equation 4.15.2 [M2]:
 = $Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 = $90687 \cdot 676/4 ((1 + 2(81.6^2 - 0^2)/(676^2))/(1 + (4 \cdot 0)/(3 \cdot 676)) - 4 \cdot 75.5/676)$
 = 89143.0 kgf-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 = $0 \cdot 816/(2 \cdot 115) - 89143/(\pi \cdot 816^2 \cdot 115)$
 = -37.10 kgf/cm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 = $0 \cdot 816/(2 \cdot 115) + 89143/(\pi \cdot 816^2 \cdot 115)$
 = 37.10 kgf/cm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 = $P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$
 = $0 \cdot 816/(2 \cdot 115) - 3189/(0.11 \cdot \pi \cdot 816^2 \cdot 115)$
 = 12.45 kgf/cm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 = $P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$
 = $0 \cdot 816/(2 \cdot 115) + 3189/(0.19 \cdot \pi \cdot 816^2 \cdot 115)$
 = -6.90 kgf/cm²

Maximum Shear Force in the Saddle (4.15.3) [T]:
 = $Q(L - 2a)/(L + (4 \cdot hm/3))$

$$= 90687 (676 - 2 \cdot 75.5) / (676 + (4 \cdot 0/3))$$

$$= 70414.8 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 70415 / (816 \cdot 115)$$

$$= 87.90 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(816 \cdot 115)}$$

$$= 238.867 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250 + 1.56 \cdot \sqrt{(816 \cdot 115)}, 2 \cdot 75.5)$$

$$= 727.73 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 90687 \cdot 0.1 / (728(115 + 1 \cdot 24))$$

$$= -6.82 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -90687 / (4(115 + 1 \cdot 24)728) -$$

$$3 \cdot 0.047 \cdot 90687 / (2(115 + 1 \cdot 24)^2)$$

$$= -55.48 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1500.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(758 \cdot 115)} + 740)115^3) / 12$$

$$= 15216.129 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 15216 + 1381(107 - 57.5)^2$$

$$= 48625.980 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1200.6	115.0	57.5	1380.7	7938857.0	0.486E+05
Wearplate	740.0	24.0	127.0	177.6	2255520.0	818.
Web	24.0	363.0	320.5	87.1	2792196.5	0.494E+05
BasePlate	275.0	40.0	522.0	110.0	5742000.0	0.190E+06
Totals	...	...	...	1755.4	18728574.0	0.289E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18728574 / 1755$$

$$= 106.692 \text{ mm}$$

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120/2
 = 120.0

Saddle Splitting Coefficient [K1]:
 = (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
 = (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))
 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.2 • 90687
 = 18456.7285 kgf

Tension Stress, St = (Fh/As) = 49.2548 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1590.6736 kgf/cm²

Saddle Splitting Dimension [d]:
 = B - R • sin(theta/2)/(theta/2 in radians)
 = 1300 - 758 • sin(120/2)/1.05
 = 673.139 mm

Bending Moment, M = Fh * d = 12423.9580 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 45.9120 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1767.4150 kgf/cm²

Minimum Thickness of Baseplate per Moss:

= $\sqrt{\frac{3(Q + \text{SaddleWt})\text{BasePlateWidth}}{(4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}}$
 = $\sqrt{\frac{3(90687 + 754)275}{(4 \cdot 1500 \cdot 1767)}}$
 = 26.672 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120/2)(755 + 118 + 24)
 = 1553.650 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 1554/(5 - 1)
 = 388.412 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 388 • 275/2
 = 534.067 cm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 91441/(1500 • 275)
 = 22.167 kgf/cm²

Axial Load [P]:
 = Ap • Bp
 = 534 • 22.2
 = 11838.921 kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 54.2 + 46.6
 = 100.849 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 11839/101
 = 117.392 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	250.0	...	60.0	...	0.312E+04

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 250 - 24 \\
 &= 226.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0/101 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 250/2 \\
 &= 125.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(3125/101)} \\
 &= 55.666 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1300 - \cos(120/2)(755 + 118 + 24) - 40 \\
 &= 811.500 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120/2)(755 + 118 + 24) \\
 &= 1553.650 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 1500 - 50.8 \\
 &= 1449.200 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (1554 - 1449) / 2 \\
 &= 52.225 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{Lv^2 + Dd^2} \\
 &= \sqrt{812^2 + 52.2^2} \\
 &= 813.179 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2651)} \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 813 / 55.7 \\
 &= 14.608
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 48789 / (2 \cdot 1500) \cdot 388 \cdot 813 / 2 \\
 &= 2568.334 \text{ kgf-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($14.6 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.6)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (14.6) / (8 \cdot 123) - (14.6^3) / (8 \cdot 123^3))$$

$$= 1539 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 117/1539 + (2568 \cdot 125/31250000) / 1767$$

$$= 0.658$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	226.0	0.0	54.2	0.0	0.312E+04
Web	388.4	24.0	0.0	93.2	0.0	44.7
Totals	...	...	...	147.5	...	0.317E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/147$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250/2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1300 - \cos(897/2) (40 + 0 + 0) - 0$$

$$= 363.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3167/147)}$$

$$= 46.343 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 363/46.3$$

$$= 7.833$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 48789 / (2 \cdot 1500)$$

$$= 16.263 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 16.3 \cdot 388 \cdot 363$$

$$= 2292.990 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2293/253358$$

$$= 905.038 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ ($7.83 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.83)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (7.83) / (8 \cdot 123) - (7.83^3) / (8 \cdot 123^3))$$

$$= 1565 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 161/1565 + (2293 \cdot 125/3167) / 1767$$

$$= 0.615$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.92 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BlArea	14.0743 cm ²
Saddle Load QO (Weight)	QO	91440.9 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	48789.1 kgf
Maximum Transverse Force	Ft	0.0 kgf
Maximum Longitudinal Force	Fl	48789.1 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 48789 / (14.1 \cdot 4)$$

$$= 866.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (14.1 \cdot 4)$$

$$= 0.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 48789 / (1052 • 4)
 = 11.5905 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1300 • 0 + 0
 = 0.00 kgf-m.

Eccentricity (e):
 = Rmom / QO
 = 0/91441
 = 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 11.6, 0]
 = 11.5905 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

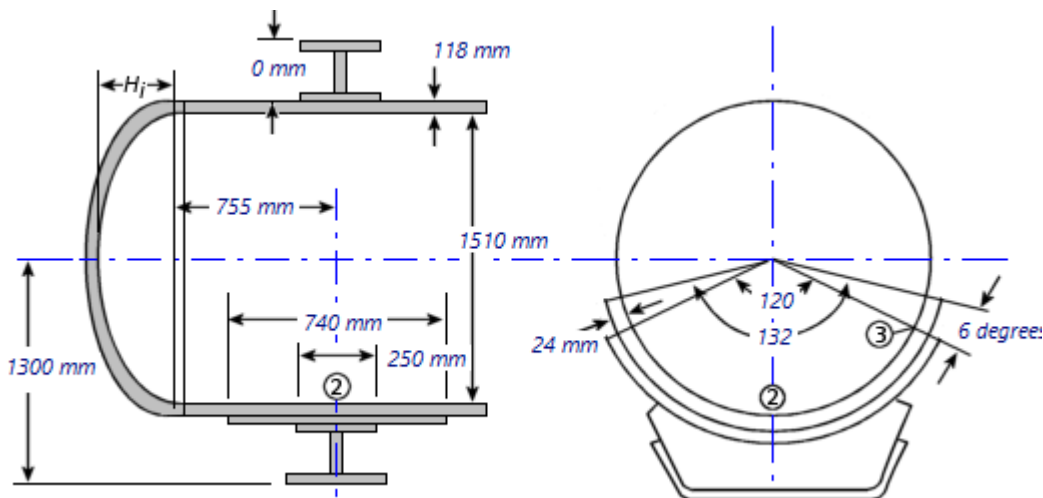
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	815.50	mm
Shell Thickness used in this Case	t	118.000	mm
Stiffened Vessel Length per 4.15.6	L	675.50	cm
Distance from Saddle to Vessel tangent	a1 or a	755.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	740.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	121192.21	kgf
Pressure used in Analysis	P 194.774	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	641.02	1407.13
Long. Stress at Bottom of Midspan	740.18	1407.13
Long. Stress at Top of Saddles	707.24	1407.13
Long. Stress at Bottom of Saddles	681.38	1407.13
Tangential Shear in Shell	117.47	1125.70
Circ. Stress at Horn of Saddle	74.15	1758.91
Circ. Compressive Stress in Shell	9.11	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (2010/2 + 0) \cdot 1300/1500$$

$$= 2612.9 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(614, 0) \cdot 1300/5245$$

$$= 152.1 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(17855, 0) \cdot 1300/5245$$

$$= 4425.4 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (17855/2 + 0) \cdot 1300/1500$$

$$= 23211.4 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 97981 + \max(152, 2613, 4425, 23211)$$

$$= 121192.2 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (0 + \pi/4(1.75 \cdot 1.2)^2)$$

$$= 6019.491 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	121946.30	kgf
Transverse Shear Load	8927.47	kgf
Longitudinal Shear Load	17854.94	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1500.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0470	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0386			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -121192 \cdot 75.5 [1 - (1 - 75.5/676 + (81.6^2 - 0^2)/(2 \cdot 75.5 \cdot 676))/(1 + (4 \cdot 0)/(3 \cdot 676))]$$

$$= -4261.1 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 121192 \cdot 676/4((1 + 2(81.6^2 - 0^2)/(676^2))/(1 + (4 \cdot 0)/(3 \cdot 676)) - 4 \cdot 75.5/676)$$

$$= 119129.2 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 195 \cdot 816/(2 \cdot 115) - 119129/(\pi \cdot 816^2 \cdot 115)$$

$$= 641.02 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 195 \cdot 816/(2 \cdot 115) + 119129/(\pi \cdot 816^2 \cdot 115)$$

$$= 740.18 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 195 \cdot 816 / (2 \cdot 115) - 4261 / (0.11 \cdot \pi \cdot 816^2 \cdot 115)$$

$$= 707.24 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 195 \cdot 816 / (2 \cdot 115) + 4261 / (0.19 \cdot \pi \cdot 816^2 \cdot 115)$$

$$= 681.38 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 121192 (676 - 2 \cdot 75.5) / (676 + (4 \cdot 0/3))$$

$$= 94101.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 94101 / (816 \cdot 115)$$

$$= 117.47 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(816 \cdot 115)}$$

$$= 238.867 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250 + 1.56 \cdot \sqrt{(816 \cdot 115)}, 2 \cdot 75.5)$$

$$= 727.73 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1407 / 1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 121192 \cdot 0.1 / (728 (115 + 1 \cdot 24))$$

$$= -9.11 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -121192 / (4(115 + 1 \cdot 24)728) -$$

$$3 \cdot 0.047 \cdot 121192 / (2(115 + 1 \cdot 24)^2)$$

$$= -74.15 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 5245.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12750E-04 \cdot 5245 (200 - 21.1)$$

$$= 11.963 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1500.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2450.6	kgf/cm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(758 \cdot 115)} + 740)^3) / 12$$

$$= 15216.129 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 15216 + 1381 (107 - 57.5)^2$$

$$= 48625.980 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1200.6	115.0	57.5	1380.7	7938857.0	0.486E+05
Wearplate	740.0	24.0	127.0	177.6	2255520.0	818.
Web	24.0	363.0	320.5	87.1	2792196.5	0.494E+05
BasePlate	275.0	40.0	522.0	110.0	5742000.0	0.190E+06
Totals	...	...	...	1755.4	18728574.0	0.289E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18728574 / 1755$$

$$= 106.692 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 121192$$

$$= 24665.2441 \text{ kgf}$$

Tension Stress, St = (Fh/As) = 65.8232 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1470.3536 kgf/cm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1300 - 758 \cdot \sin(120/2) / 1.05$$

$$= 673.139 \text{ mm}$$

Bending Moment, M = Fh * d = 16603.1562 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 61.3560 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1633.7263 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(121192 + 754)275) / (4 \cdot 1500 \cdot 1634)}$$

$$= 32.037 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (755 + 118 + 24)$$

$$= 1553.650 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1554 / (5 - 1)$$

$$= 388.412 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 388 \cdot 275 / 2$$

$$= 534.067 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 121946 / (1500 \cdot 275)$$

$$= 29.563 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 534 \cdot 29.6$$

$$= 15788.483 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 54.2 + 46.6$$

$$= 100.849 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 15788 / 101$$

$$= 156.555 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	250.0	...	60.0	...	0.312E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 250 - 24$$

$$= 226.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 101$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 250 / 2$$

$$= 125.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (3125 / 101) }$$

$$= 55.666 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) - bpthk$$

$$= 1300 - \cos(120 / 2)(755 + 118 + 24) - 40$$

$$= 811.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(120 / 2)(755 + 118 + 24)$$

$$= 1553.650 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 1500 - 50.8$$

$$= 1449.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (1554 - 1449) / 2$$

$$= 52.225 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{812^2 + 52.2^2}$$

$$= 813.179 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2451)}$$

$$= 128.153$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 813 / 55.7$$

$$= 14.608$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 17855 / (2 \cdot 1500) \cdot 388 \cdot 813 / 2$$

$$= 939.911 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($14.6 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.6)^2 / (2 \cdot 128^2)) 2451 /$$

$$(5/3 + 3 \cdot (14.6) / (8 \cdot 128) - (14.6)^3 / (8 \cdot 128^3))$$

$$= 1424 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 157 / 1424 + (940 \cdot 125 / 31250000) / 1634$$

$$= 0.340$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	226.0	0.0	54.2	0.0	0.312E+04
Web	388.4	24.0	0.0	93.2	0.0	44.7
Totals	...	...	...	147.5	...	0.317E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 147$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250 / 2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1300 - \cos(897/2) (40 + 0 + 0) - 0$$

$$= 363.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3167 / 147)}$$

$$= 46.343 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 363 / 46.3$$

$$= 7.833$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 17855 / (2 \cdot 1500)$$

$$= 5.952 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 5.95 \cdot 388 \cdot 363$$

$$= 839.146 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 839 / 253358$$

$$= 331.209 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.83 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.83)^2 / (2 \cdot 128^2)) 2451 /$$

$$(5/3 + 3 \cdot (7.83) / (8 \cdot 128) - (7.83^3) / (8 \cdot 128^3))$$

$$= 1448 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 214 / 1448 + (839 \cdot 125 / 3167) / 1634$$

$$= 0.351$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1529.00 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	14.0743 cm ²
Saddle Load QO (Weight)	QO	98734.9 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	4425.4 kgf
Maximum Transverse Force	Ft	8927.5 kgf
Maximum Longitudinal Force	Fl	17854.9 kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 7462.0 \text{ kgf}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 17855 / (14.1 \cdot 4)$$

$$= 317.2 \text{ kgf/cm}^2. \text{ Must be less than } 1019.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8927 / (14.1 \cdot 4)$$

$$= 158.6 \text{ kgf/cm}^2. \text{ Must be less than } 1019.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 17855 / (1019 \cdot 4)$$

$$= 4.3805 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1300 \cdot 8927 + 0$$

= 11605.72 kgf-m.

Eccentricity (e):
 = Rmom / QO
 = 11606/98735
 = 117.54 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 4.38, 0]
 = 4.3805 cm²

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	815.50	mm
Shell Thickness used in this Case	t	118.000	mm
Stiffened Vessel Length per 4.15.6	L	675.50	cm
Distance from Saddle to Vessel tangent	a1 or a	755.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	740.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.08	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		115731.97	kgf
Pressure used in Analysis	P	194.774	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		643.25	1407.13
Long. Stress at Bottom of Midspan		737.95	1407.13
Long. Stress at Top of Saddles		706.49	1407.13
Long. Stress at Bottom of Saddles		681.80	1407.13
Tangential Shear in Shell		112.17	1125.70
Circ. Stress at Horn of Saddle		70.81	1758.91
Circ. Compressive Stress in Shell		8.70	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3 (2010/2 + 0) • 1300/1500
 = 2612.9 kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(F_{l,wind}; Sum of X Forces) • B / Ls

$$= \max(614, 0) \cdot 1300/5245$$

$$= 152.1 \text{ kgf}$$

Saddle Reaction Force due to Earthquake F_I [F_{sl}]:

$$= \max(F_{I,eq}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(17855, 0) \cdot 1300/5245$$

$$= 4425.4 \text{ kgf}$$

Saddle Reaction Force due to Earthquake F_t [F_{st}]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (17855/2 + 0) \cdot 1300/1500$$

$$= 23211.4 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 92521 + \max(152, 2613, 4425, 23211)$$

$$= 115732.0 \text{ kgf}$$

Longitudinal Wind Force [F_l]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (0 + \pi/4 (1.75 \cdot 1.2)^2)$$

$$= 6019.491 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	116486.04	kgf
Transverse Shear Load	8927.47	kgf
Longitudinal Shear Load	17854.94	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1500.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0470	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0386			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -115732 \cdot 75.5 [1 - (1 - 75.5/676 + (81.6^2 - 0^2)/(2 \cdot 75.5 \cdot 676)) / (1 + (4 \cdot 0)/(3 \cdot 676))]$$

$$= -4069.1 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4h m^2)/(3L)) - 4a/L)$$

$$= 115732 \cdot 676/4 ((1 + 2(81.6^2 - 0^2)/(676^2)) / (1 + (4 \cdot 0)/(3 \cdot 676)) - 4 \cdot 75.5/676)$$

$$= 113761.9 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 195 \cdot 816 / (2 \cdot 115) - 113762 / (\pi \cdot 816^2 \cdot 115)$$

$$= 643.25 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 195 \cdot 816 / (2 \cdot 115) + 113762 / (\pi \cdot 816^2 \cdot 115)$$

$$= 737.95 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 195 \cdot 816 / (2 \cdot 115) - 4069 / (0.11 \cdot \pi \cdot 816^2 \cdot 115)$$

$$= 706.49 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 195 \cdot 816 / (2 \cdot 115) + 4069 / (0.19 \cdot \pi \cdot 816^2 \cdot 115)$$

$$= 681.80 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 115732 (676 - 2 \cdot 75.5) / (676 + (4 \cdot 0/3))$$

$$= 89861.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 89861 / (816 \cdot 115)$$

$$= 112.17 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(816 \cdot 115)}$$

$$= 238.867 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250 + 1.56 \cdot \sqrt{(816 \cdot 115)}, 2 \cdot 75.5)$$

$$= 727.73 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1407 / 1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot t_r))$$

$$= -0.76 \cdot 115732 \cdot 0.1 / (728 (115 + 1 \cdot 24))$$

$$= -8.70 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot t_r)^2)$$

$$= -115732 / (4(115 + 1 \cdot 24) 728) -$$

$$3 \cdot 0.047 \cdot 115732 / (2(115 + 1 \cdot 24)^2)$$

$$= -70.81 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1500.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2450.6	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.080	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(758 \cdot 115)} + 740) 115^3) / 12$$

$$= 15216.129 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 15216 + 1381 (107 - 57.5)^2$$

$$= 48625.980 \text{ cm}^4$$

B	D	Y	A	AY	I + AD ² (2)
mm	mm	mm	cm ²	mm ³	cm ⁴

Shell	1200.6	115.0	57.5	1380.7	7938857.0	0.486E+05
Wearplate	740.0	24.0	127.0	177.6	2255520.0	818.
Web	24.0	363.0	320.5	87.1	2792196.5	0.494E+05
BasePlate	275.0	40.0	522.0	110.0	5742000.0	0.190E+06
Totals	...	...	...	1755.4	18728574.0	0.289E+06

Distance to Centroid [C1]:
 = AY / A
 = $18728574/1755$
 = 106.692 mm

Angle [β]:
 = $180 - \text{Saddle Angle}/2$
 = $180 - 120/2$
 = 120.0

Saddle Splitting Coefficient [K1]:
 = $(1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 = $(1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 = 0.2035

Saddle Splitting Force [Fh]:
 = $K1 \cdot Q$
 = $0.2 \cdot 115732$
 = 23553.9648 kgf

Tension Stress, $St = (Fh/As) = 62.8576 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 1470.3536 \text{ kgf/cm}^2$

Saddle Splitting Dimension [d]:
 = $B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 = $1300 - 758 \cdot \sin(120/2) / 1.05$
 = 673.139 mm

Bending Moment, $M = Fh \cdot d = 15855.1104 \text{ kgf-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 58.5916 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 1633.7263 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

= $\sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 = $\sqrt{3(115732 + 754) \cdot 275 / (4 \cdot 1500 \cdot 1634)}$
 = 31.311 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = $2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 = $2 \cdot \cos(90 - 120/2) (755 + 118 + 24)$
 = 1553.650 mm

Distance between Ribs [e]:
 = $\text{Web Length} / (\text{Nribs} - 1)$
 = $1554 / (5 - 1)$
 = 388.412 mm

Baseplate Pressure Area [Ap]:
 = $e \cdot \text{Bpwid} / 2$
 = $388 \cdot 275 / 2$
 = 534.067 cm²

Bearing Pressure [Bp]:
 = $(Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 = $116486 / (1500 \cdot 275)$
 = 28.239 kgf/cm²

Axial Load [P]:
 = $Ap \cdot Bp$
 = $534 \cdot 28.2$

= 15081.541 kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 54.2 + 46.6
 = 100.849 cm²

Compressive Stress [Sc]:

= P/Ar
 = 15082/101
 = 149.545 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	250.0	...	60.0	...	0.312E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 250 - 24
 = 226.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/101
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 250/2
 = 125.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{(3125/101)}$
 = 55.666 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 1300 - cos(120/2)(755 + 118 + 24) - 40
 = 811.500 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(120/2)(755 + 118 + 24)
 = 1553.650 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance
 = 1500 - 50.8
 = 1449.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (1554 - 1449)/2
 = 52.225 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$
 = $\sqrt{812^2 + 52.2^2}$
 = 813.179 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2038900/2451}$
 = 128.153

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 813/55.7$$

$$= 14.608$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 17855 / (2 \cdot 1500) \cdot 388 \cdot 813 / 2$$

$$= 939.911 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($14.6 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.6)^2 / (2 \cdot 128^2)) 2451 / (5/3 + 3 \cdot (14.6) / (8 \cdot 128) - (14.6^3) / (8 \cdot 128^3))$$

$$= 1424 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 150/1424 + (940 \cdot 125/31250000) / 1634$$

$$= 0.335$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	226.0	0.0	54.2	0.0	0.312E+04
Web	388.4	24.0	0.0	93.2	0.0	44.7
Totals	...	...	...	147.5	...	0.317E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/147$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250/2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1300 - \cos(897/2) (40 + 0 + 0) - 0$$

$$= 363.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (3167/147) }$$

$$= 46.343 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 363/46.3$$

$$= 7.833$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 17855 / (2 \cdot 1500)$$

$$= 5.952 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 5.95 \cdot 388 \cdot 363$$

$$= 839.146 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 839/253358$$

$$= 331.209 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.83 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (K/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K/r) / (8 \cdot C_c) - (K/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.83)^2 / (2 \cdot 128^2)) 2451 /$$

$$(5/3 + 3 \cdot (7.83) / (8 \cdot 128) - (7.83^3) / (8 \cdot 128^3))$$

$$= 1448 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 205 / 1448 + (839 \cdot 125 / 3167) / 1634$$

$$= 0.344$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1529.00 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BlArea	14.0743 cm ²
Saddle Load Q0 (Weight)	Q0	93274.6 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	4425.4 kgf
Maximum Transverse Force	Ft	8927.5 kgf
Maximum Longitudinal Force	Fl	17854.9 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 17855 / (14.1 \cdot 4)$$

$$= 317.2 \text{ kgf/cm}^2. \text{ Must be less than } 1019.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8927 / (14.1 \cdot 4)$$

$$= 158.6 \text{ kgf/cm}^2. \text{ Must be less than } 1019.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 17855 / (1019 \cdot 4)$$

$$= 4.3805 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1300 \cdot 8927 + 0$$

$$= 11605.72 \text{ kgf-m.}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$

$$= 11606 / 93275$$

$$= 124.43 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 4.38, 0]$$

$$= 4.3805 \text{ cm}^2$$



Note: Saddle and wear plate material are both specifed as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

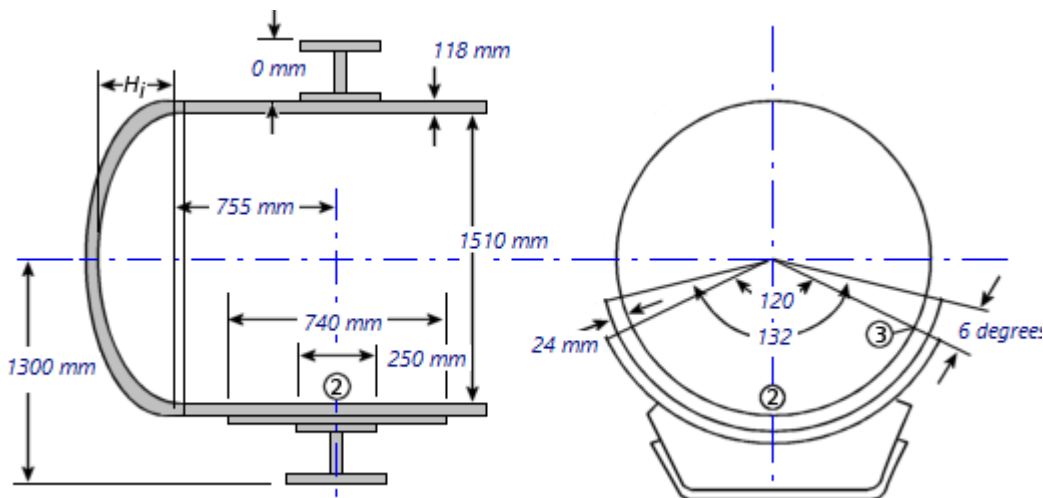
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	815.50	mm
Shell Thickness used in this Case	t	118.000	mm
Stiffened Vessel Length per 4.15.6	L	675.50	cm
Distance from Saddle to Vessel tangent	a1 or a	755.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	740.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		99720.77	kgf
Pressure used in Analysis	P	248.522	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		840.38	2518.57
Long. Stress at Bottom of Midspan		921.97	2518.57
Long. Stress at Top of Saddles		894.86	2518.57
Long. Stress at Bottom of Saddles		873.59	2518.57
Tangential Shear in Shell		96.66	1413.05
Circ. Stress at Horn of Saddle		61.01	3777.85
Circ. Compressive Stress in Shell		7.49	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$\begin{aligned}
 &= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E \\
 &= 3 (663/2 + 0) \cdot 1300/1500 \\
 &= 862.3 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Longitudinal Saddle Reaction Force [Fwl]:} \\
 &= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls \\
 &= \max(203, 0) \cdot 1300/5245 \\
 &= 50.2 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Load Combination Results for Q + Wind or Seismic [Q]:} \\
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= 98858 + \max(50.2, 862, 0, 0) \\
 &= 99720.8 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Longitudinal Wind Force [FI]:} \\
 &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 1587 (0 + \pi/4(1.75 \cdot 1.2)^2) \\
 &= 6019.491 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	100474.84	kgf
Transverse Shear Load	331.64	kgf
Longitudinal Shear Load	202.56	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

$$\begin{aligned}
 &\text{Saddle Dimension [E]:} \\
 &= \text{Baseplate Length} \\
 &= 1500.000 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0470	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0386			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

$$\begin{aligned}
 &\text{Moment per Equation 4.15.1 [M1]:} \\
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 \cdot hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -99721 \cdot 75.5 [1 - (1 - 75.5/676 + (81.6^2 \cdot 0^2)/(2 \cdot 75.5 \cdot 676))/(1 + (4 \cdot 0)/(3 \cdot 676))] \\
 &= -3506.2 \text{ kgf-m.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Moment per Equation 4.15.2 [M2]:} \\
 &= Q \cdot L/4((1 + 2(Rm^2 \cdot hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 99721 \cdot 676/4((1 + 2(81.6^2 \cdot 0^2)/(676^2))/(1 + (4 \cdot 0)/(3 \cdot 676)) - 4 \cdot 75.5/676) \\
 &= 98023.2 \text{ kgf-m.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Longitudinal Stress at Top of Shell (4.15.4) [\sigma1]:} \\
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 249 \cdot 816/(2 \cdot 115) - 98023/(\pi \cdot 816^2 \cdot 115) \\
 &= 840.38 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Longitudinal Stress at Bottom of Shell (4.15.5) [\sigma2]:} \\
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 249 \cdot 816/(2 \cdot 115) + 98023/(\pi \cdot 816^2 \cdot 115) \\
 &= 921.97 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Longitudinal Stress at Top of Shell at Support (4.15.8) [\sigma*3]:} \\
 &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 249 \cdot 816/(2 \cdot 115) - 3506/(0.11 \cdot \pi \cdot 816^2 \cdot 115) \\
 &= 894.86 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Longitudinal Stress at Bottom of Shell at Support (4.15.9) [\sigma*4]:} \\
 &= P \cdot Rm/(2t) + M1/(K1* \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 249 \cdot 816/(2 \cdot 115) + 3506/(0.19 \cdot \pi \cdot 816^2 \cdot 115)
 \end{aligned}$$

$$= 873.59 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 99721 (676 - 2 \cdot 75.5)/(676 + (4 \cdot 0/3))$$

$$= 77429.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 77429 / (816 \cdot 115)$$

$$= 96.66 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(816 \cdot 115)}$$

$$= 238.867 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250 + 1.56 \cdot \sqrt{(816 \cdot 115)}, 2 \cdot 75.5)$$

$$= 727.73 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 99721 \cdot 0.1 / (728 (115 + 1 \cdot 24))$$

$$= -7.49 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q/(4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q/(2(t + \eta \cdot tr)^2)$$

$$= -99721/(4(115 + 1 \cdot 24)728) -$$

$$3 \cdot 0.047 \cdot 99721/(2(115 + 1 \cdot 24)^2)$$

$$= -61.01 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1500.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(758 \cdot 115)} + 740)115^3)/12$$

$$= 15216.129 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 15216 + 1381 (107 - 57.5)^2$$

$$= 48625.980 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1200.6	115.0	57.5	1380.7	7938857.0	0.486E+05
Wearplate	740.0	24.0	127.0	177.6	2255520.0	818.
Web	24.0	363.0	320.5	87.1	2792196.5	0.494E+05
BasePlate	275.0	40.0	522.0	110.0	5742000.0	0.190E+06
Totals	...	...	...	1755.4	18728574.0	0.289E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18728574/1755$$

$$= 106.692 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 99721$$

$$= 20295.3379 \text{ kgf}$$

Tension Stress, $St = (Fh/As) = 54.1614 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1300 - 758 \cdot \sin(120/2) / 1.05$$

$$= 673.139 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 13661.5996 \text{ kgf-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 50.4856 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(99721 + 754) \cdot 275 / (4 \cdot 1500 \cdot 1767)}$$

$$= 27.958 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (755 + 118 + 24)$$

$$= 1553.650 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1554 / (5 - 1)$$

$$= 388.412 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 388 \cdot 275 / 2$$

$$= 534.067 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 100475 / (1500 \cdot 275)$$

$$= 24.358 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 534 \cdot 24.4$$

$$= 13008.558 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 54.2 + 46.6$$

$$= 100.849 \text{ cm}^2$$

Compressive Stress [Sc]:

= P/Ar
 = 13009/101
 = 128.990 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	250.0	...	60.0	...	0.312E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 250 - 24
 = 226.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/101
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 250/2
 = 125.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{3125/101}$
 = 55.666 mm

Length of Outer Rib, step 1, [Lv]:
 = Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 1300 - cos(120/2)(755 + 118 + 24) - 40
 = 811.500 mm

Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(120/2) (755 + 118 + 24)$
 = 1553.650 mm

Length of Outer Rib, step 3, [Bd]:
 = Baseplate length - clearance
 = 1500 - 50.8
 = 1449.200 mm

Length of Outer Rib, step 4, [Dd]:
 $= (Lw - Bd) / 2$
 $= (1554 - 1449) / 2$
 = 52.225 mm

Length of Outer Rib [L]:
 $= \sqrt{Lv^2 + Dd^2}$
 $= \sqrt{812^2 + 52.2^2}$
 = 813.179 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 2038900 / 2651}$
 = 123.211

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 813/55.7
 = 14.608

Bending Moment [Rm]:
 $= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$
 $= 203 / (2 \cdot 1500) \cdot 388 \cdot 813 / 2$

= 10.663 kgf-m.

Compressive Allowable, $KL/r < Cc$ (14.6 < 123) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(KL/r)^2}{(2 \cdot Cc^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot Cc)} - \frac{(KL/r)^3}{(8 \cdot Cc^3)} \right)$$

$$= \left(1 - \frac{(14.6)^2}{(2 \cdot 123^2)} \right) 2651 / \left(\frac{5}{3} + 3 \cdot \frac{(14.6)}{(8 \cdot 123)} - \frac{(14.6^3)}{(8 \cdot 123^3)} \right)$$

$$= 1539 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 129/1539 + (10.7 \cdot 125/31250000) / 1767$$

$$= 0.086$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	226.0	0.0	54.2	0.0	0.312E+04
Web	388.4	24.0	0.0	93.2	0.0	44.7
Totals	...	...	...	147.5	...	0.317E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/147$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250/2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1300 - \cos(897/2) (40 + 0 + 0) - 0$$

$$= 363.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 3167/147 }$$

$$= 46.343 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 363/46.3$$

$$= 7.833$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 203 / (2 \cdot 1500)$$

$$= 0.068 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.068 \cdot 388 \cdot 363$$

$$= 9.520 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9.52/253358$$

$$= 3.757 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ (7.83 < 123) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(KL/r)^2}{(2 \cdot Cc^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot Cc)} - \frac{(KL/r)^3}{(8 \cdot Cc^3)} \right)$$

$$= \left(1 - \frac{(7.83)^2}{(2 \cdot 123^2)} \right) 2651 / \left(\frac{5}{3} + 3 \cdot \frac{(7.83)}{(8 \cdot 123)} - \frac{(7.83^3)}{(8 \cdot 123^3)} \right)$$

$$= 1565 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 177/1565 + (9.52 \cdot 125/3167)/1767$$

$$= 0.115$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.92 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	14.0743 cm ²
Saddle Load QO (Weight)	QO	99612.6 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	50.2 kgf
Maximum Transverse Force	Ft	331.6 kgf
Maximum Longitudinal Force	Fl	202.6 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 203 / (14.1 \cdot 4)$
 $= 3.6 \text{ kgf/cm}^2$. Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 332 / (14.1 \cdot 4)$
 $= 5.9 \text{ kgf/cm}^2$. Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 203 / (1052 \cdot 4)$
 $= 0.0481 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1300 \cdot 332 + 0$
 $= 431.13 \text{ kgf-m}$.

Eccentricity (e):
 $= Rmom / QO$
 $= 431/99613$
 $= 4.33 \text{ mm} < Bplen/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0, 0.048, 0]$
 $= 0.0481 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	815.50	mm
Shell Thickness used in this Case	t	118.000	mm
Stiffened Vessel Length per 4.15.6	L	675.50	cm
Distance from Saddle to Vessel tangent	a1 or a	755.00	mm

Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	740.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.08	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		94220.28	kgf
Pressure used in Analysis	P	248.522	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	842.63	2518.57
Long. Stress at Bottom of Midspan	919.72	2518.57
Long. Stress at Top of Saddles	894.11	2518.57
Long. Stress at Bottom of Saddles	874.00	2518.57
Tangential Shear in Shell	91.32	1413.05
Circ. Stress at Horn of Saddle	57.64	3777.85
Circ. Compressive Stress in Shell	7.08	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (663/2 + 0) \cdot 1300/1500$
 $= 862.3 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(203, 0) \cdot 1300/5245$
 $= 50.2 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 93358 + \max(50.2, 862, 0, 0)$
 $= 94220.3 \text{ kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4 (1.75 \cdot 1.2)^2)$
 $= 6019.491 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	94974.35	kgf
Transverse Shear Load	331.64	kgf
Longitudinal Shear Load	202.56	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 1500.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0470	K8 = 0.3405



K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0386

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -94220 \cdot 75.5 [1 - (1 - 75.5/676 + (81.6^2 - 0^2)/(2 \cdot 75.5 \cdot 676)) / (1 + (4 \cdot 0)/(3 \cdot 676))]$$

$$= -3312.8 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 94220 \cdot 676/4 ((1 + 2(81.6^2 - 0^2)/(676^2)) / (1 + (4 \cdot 0)/(3 \cdot 676)) - 4 \cdot 75.5/676)$$

$$= 92616.4 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 249 \cdot 816/(2 \cdot 115) - 92616/(\pi \cdot 816^2 \cdot 115)$$

$$= 842.63 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 249 \cdot 816/(2 \cdot 115) + 92616/(\pi \cdot 816^2 \cdot 115)$$

$$= 919.72 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 249 \cdot 816/(2 \cdot 115) - 3313/(0.11 \cdot \pi \cdot 816^2 \cdot 115)$$

$$= 894.11 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 249 \cdot 816/(2 \cdot 115) + 3313/(0.19 \cdot \pi \cdot 816^2 \cdot 115)$$

$$= 874.00 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 94220 (676 - 2 \cdot 75.5)/(676 + (4 \cdot 0/3))$$

$$= 73158.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 73158/(816 \cdot 115)$$

$$= 91.32 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(816 \cdot 115)}$$

$$= 238.867 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250 + 1.56 \cdot \sqrt{(816 \cdot 115)}, 2 \cdot 75.5)$$

$$= 727.73 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000$$
 Materials are the same, test case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 94220 \cdot 0.1/(728(115 + 1 \cdot 24))$$

$$= -7.08 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q/(4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q/(2(t + \eta \cdot tr)^2)$$

$$= -94220/(4(115 + 1 \cdot 24)728) -$$

$$3 \cdot 0.047 \cdot 94220 / (2(115 + 1 \cdot 24)^2) \\ = -57.64 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1500.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.080	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12 \\ = ((1.56 \cdot \sqrt{(758 \cdot 115)} + 740)^3) / 12 \\ = 15216.129 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2 \\ = 15216 + 1381 (107 - 57.5)^2 \\ = 48625.980 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1200.6	115.0	57.5	1380.7	7938857.0	0.486E+05
Wearplate	740.0	24.0	127.0	177.6	2255520.0	818.
Web	24.0	363.0	320.5	87.1	2792196.5	0.494E+05
BasePlate	275.0	40.0	522.0	110.0	5742000.0	0.190E+06
Totals	...	...	...	1755.4	18728574.0	0.289E+06

Distance to Centroid [C1]:

$$= AY / A \\ = 18728574 / 1755 \\ = 106.692 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2 \\ = 180 - 120 / 2 \\ = 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ = (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ = 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q \\ = 0.2 \cdot 94220 \\ = 19175.8691 \text{ kgf}$$

Tension Stress, St = (Fh/As) = 51.1739 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1590.6736 kgf/cm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ = 1300 - 758 \cdot \sin(120/2) / 1.05 \\ = 673.139 \text{ mm}$$

Bending Moment, M = Fh * d = 12908.0410 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 47.7009 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1767.4150 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(94220 + 754) \cdot 275 / (4 \cdot 1500 \cdot 1767))}$$

$$= 27.182 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (755 + 118 + 24)$$

$$= 1553.650 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1554 / (5 - 1)$$

$$= 388.412 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 388 \cdot 275 / 2$$

$$= 534.067 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 94974 / (1500 \cdot 275)$$

$$= 23.024 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 534 \cdot 23$$

$$= 12296.405 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 54.2 + 46.6$$

$$= 100.849 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 12296 / 101$$

$$= 121.928 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	250.0	...	60.0	...	0.312E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 250 - 24$$

$$= 226.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 101$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250 / 2$$

$$= 125.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3125 / 101)}$$

$$= 55.666 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:
 = Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 1300 - cos(120/2)(755 + 118 + 24) - 40
 = 811.500 mm

Length of Outer Rib, step 2, [Lw]:
 = 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(120/2)(755 + 118 + 24)
 = 1553.650 mm

Length of Outer Rib, step 3, [Bd]:
 = Baseplate length - clearance
 = 1500 - 50.8
 = 1449.200 mm

Length of Outer Rib, step 4, [Dd]:
 = (Lw - Bd)/2
 = (1554 - 1449)/2
 = 52.225 mm

Length of Outer Rib [L]:
 = $\sqrt{(Lv^2 + Dd^2)}$
 = $\sqrt{(812^2 + 52.2^2)}$
 = 813.179 mm

Intermediate Term [Cc]:
 = $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 2038900 / 2651)}$
 = 123.211

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 813/55.7
 = 14.608

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 203 / (2 • 1500) • 388 • 813/2
 = 10.663 kgf-m.

Compressive Allowable, KL/r < Cc (14.6 < 123) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (14.6)^2 / (2 \cdot 123^2)) 2651 / (5/3 + 3 \cdot (14.6) / (8 \cdot 123) - (14.6)^3 / (8 \cdot 123^3))$
 = 1539 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I)/Sba
 = 122/1539 + (10.7 • 125/31250000)/1767
 = 0.082

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	226.0	0.0	54.2	0.0	0.312E+04
Web	388.4	24.0	0.0	93.2	0.0	44.7
Totals	...	...	...	147.5	...	0.317E+04

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/147
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 250/2
 = 125.000 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = $1300 - \cos(897/2)(40 + 0 + 0) - 0$
 = 363.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{3167/147}$
 = 46.343 mm

Slenderness ratio [KL/r]:
 = KL/r
 = $1 \cdot 363/46.3$
 = 7.833

Unit Force [Force,u]:
 = FI / (2 • Baseplate Length)
 = $203/(2 \cdot 1500)$
 = 0.068 kgf/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 = $0.068 \cdot 388 \cdot 363$
 = 9.520 kgf-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = $9.52/253358$
 = 3.757 kgf/cm²

Compressive Allowable, KL/r < Cc (7.83 < 123) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2/(2 \cdot Cc^2))F_y/(5/3 + 3 \cdot (KL/r)/(8 \cdot Cc) - (KL/r)^3/(8 \cdot Cc^3))$
 = $(1 - (7.83)^2/(2 \cdot 123^2))2651/$
 = $(5/3 + 3 \cdot (7.83)/(8 \cdot 123) - (7.83^3)/(8 \cdot 123^3))$
 = 1565 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = Sc/Sca + (Mbase,c • C1/I)/Sba
 = $167/1565 + (9.52 \cdot 125/3167)/1767$
 = 0.109

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.92 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	14.0743 cm ²
Saddle Load Q0 (Weight)	Q0	94112.1 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	50.2 kgf
Maximum Transverse Force	Ft	331.6 kgf
Maximum Longitudinal Force	F1	202.6 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 = FI / (Bolt Area • Number of Bolts)
 = $203/(14.1 \cdot 4)$
 = 3.6 kgf/cm². Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 = Ft / (Bolt Area • Number of Bolts)
 = $332/(14.1 \cdot 4)$
 = 5.9 kgf/cm². Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss



Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
= $203 / (1052 \cdot 4)$
= 0.0481 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1300 \cdot 332 + 0$
= 431.13 kgf-m.

Eccentricity (e):
= R_{mom} / QO
= $431 / 94112$
= 4.58 mm < $B_{\text{plen}} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0, 0.048, 0]$
= 0.0481 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: T2-Tube outlet

From: 30

Pressure for Reinforcement Calculations	P	149.902	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1510.00	mm
Shell Finished (Minimum) Thickness	t	92.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		137.82	cm
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

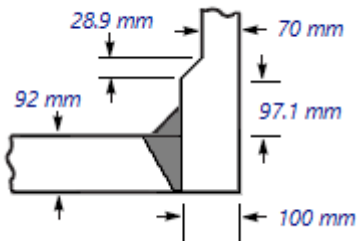
Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	70.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	97.1325	mm
Height of Beveled Transition	L'	28.8675	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	90.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T2-Tube outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	10.000	in.
Actual Thickness Used in Calculation	2.756	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 260 \cdot (\exp(150/(1407 \cdot 1)) - 1) + 3 + 0$
 $= 14.6 + 3 + 0 = 17.6 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 67 + 260) / 260) - 0$
 $= 585 - 0 = 585 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 88.293 mm

Nozzle neck minimum required thickness : 11.340 mm

Minimum nozzle neck thickness 70.002 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 257.8,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3260.0 kgf/cm ²	Passed
Occasional	: 115.6,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 123.2,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1407/1407, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 100 - 3$
 $= 97.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 70 - 3$
 $= 67.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1516/89$
 $= 17.034$ must be less than 400.



Effective Pressure Radius [Reff]:

$$= Di/2 + \text{corrosion allowance}$$

$$= 1510/2 + 3$$

$$= 758.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$$

$$= \min(\sqrt{758 \cdot 89}, 2 \cdot 130)$$

$$= 259.734 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, te] + \sqrt{\frac{Rn \cdot tn}{1}}$$

$$= \min[1.5 \cdot 89, 0] + \sqrt{130 \cdot 97}$$

$$= 112.294 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= Lpr1$$

$$= 300$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + te)$$

$$= 8(89 + 0)$$

$$= 712.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[112, 300, 712] + 89$$

$$= 201.294 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 89 + ((0 \cdot 1) / 260)$$

$$= 89.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 130 + 97) / (\sqrt{(1516 + 89) \cdot 89}))$$

$$= 0.945$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 89 \cdot 260 \cdot \max((0.945/5)^{0.85}, 1)$$

$$= 231.164 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 181 + 13.5$$

$$= 194.061 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg41^2$$

$$= 0.5 \cdot 10^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 231 + 1(194 + 0) + 0.5 + 0 + 0$$

$$= 425.725 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 97 / \ln[1 + 97/130]$$

$$= 174.017 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 89 / \ln[1 + 89/758]$$

$$= 801.677 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407 \cdot 1$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 150 \cdot 174 \cdot 201$$

$$= 52508.6 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 150 \cdot 802 (260 + 97)$$

$$= 428697.2 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 150 \cdot 802 \cdot 130$$

$$= 156224.4 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (428697 + 156224 + 52509) / 150$$

$$= 4252.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2111 / (2 \cdot 4252 / 426 - 802 / 89)$$

$$= 192.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 1407 [89/802]$$

$$= 156.2 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(192, 156)$$

$$= 156.216 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (52509 + 428697 + 156224) / 426$$

$$= 1497.285 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 150 \cdot 802 / 89$$

$$= 1350.3 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1497 - 1350, 1350)$$

$$= 1644.3 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1644.31	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	156.22	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (130 + 97) / 130$$

$$= 1.746 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (130 + 97)$$

$$= 356.571 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 97.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(156224 \cdot 1.75, 1.5 \cdot 1407(194+0), \pi/4 \cdot 150 \cdot 130^2 \cdot 1.75^2)$$

$$= 60666.352 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 60666 / (357(0.49 \cdot 7.07 + 0.6 \cdot 89 + 0.49 \cdot 0))$$

$$= 299.198 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle away from the intersecting weld by 188.258 mm.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 0 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.218

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (14.6 \cdot 1) / (70 - 3 - 0)$$

$$= 0.218$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 18 °C
User entered Minimum Design Temperature	: 0 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: 20 °C
Temperature reduction per Graph 3.13	: 2 °C
The computed stress ratio is	: 0.957

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (85.2 \cdot 1) / (92 - 3 - 0)$$

$$= 0.957$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 18 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -42 °C

The Drop for this Nozzle is : 34.9333 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.9333 mm

Input Echo, WRC297 Item 1, Description: T2-Tube outlet:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1510.000	mm	
Shell Thickness	92.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	254.000	mm	
Nozzle Thickness	100.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-266 2		
Nozzle Cold S.I. Allowable	SNmc	1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.13	kgf/cm ²

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	149.90	kgf/cm ²
Radial Load	P	-8158.12	kgf
Circumferential Shear	Vc	8158.12	kgf
Longitudinal Shear	Vl	6118.59	kgf
Circumferential Moment	Mc	5300.64	kgf-m.
Longitudinal Moment	Ml	5300.64	kgf-m.
Torsional Moment	Mt	6116.12	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (18.413) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tv, (4.935) is < 5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (4.540) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 454.0 + 2 \cdot 1.65 \cdot \sqrt{(802.5(92.0 - 3.0))}$
 $= 1335.925 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1605.000 mm
Nozzle Outside Diameter	(d) =	454.000 mm
Vessel Thickness used	(T) =	89.000 mm
Nozzle Thickness used	(t) =	97.000 mm
T / t	=	0.918
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.201$$

Nr/P	=	0.112
Mr/P	=	0.172
M0/P	=	0.057
N0/P	=	0.116
MrD/Mc	=	0.279
NrDL/Mc	=	0.137
M0d/Mc	=	0.102
N0DL/Mc	=	0.044



MrD/M1 = 0.176
 NrDL/M1 = 0.100
 MØD/M1 = 0.064
 NØDL/M1 = 0.128

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	11	11	11	11
Outplane Bending (P)	35	-35	35	-35
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-56	56	56	-56
Normal Pressure Stress	1205	1355	1205	1355
Outplane Stress Summary	1177	1369	1328	1294

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	106	-106	106	-106
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-14	-14	14	14
Inplane Bending (ML)	-155	155	155	-155
Inplane Pressure Stress	602	602	602	602
Inplane Stress Summary	550	648	890	367

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	21	21	21	21
Shear Stress Summary	33	33	8	8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1179	1371	1328	1295

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	11	11	11	11
Outplane Bending (P)	35	-35	35	-35
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-90	90	90	-90
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	602	602	602	602
Outplane Stress Summary	553	663	746	495

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	106	-106	106	-106
Inplane Membrane (Mc)	-20	-20	20	20
Inplane Bending (Mc)	-246	246	246	-246
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1205	1355	1205	1355
Inplane Stress Summary	1057	1487	1590	1034

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	21	21	21	21
Shear Stress Summary	11	11	30	30

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1057	1487	1591	1036

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	11	11	11	11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	175	355	175	355
Hoop Stress Summary	168	348	206	386

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	7	7	7	7
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-64	-64	64	64
Axial Bending (ML)	-93	93	93	-93
Axial Pressure Stress	87	87	87	87
Axial Stress Summary	-2	63	313	5

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	11	11	-11	-11
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	19	19	19	19
Shear Stress Summary	31	31	7	7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	182	351	314	386

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	11	11	11	11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	175	355	175	355
Hoop Stress Summary	181	360	193	373

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	7	7	7	7
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	-64	-64	64	64
Axial Bending (Mc)	-156	156	156	-156
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	87	87	87	87
Axial Stress Summary	-65	126	376	-56

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-8	-8	8	8
Shear due to Torsion	19	19	19	19
Shear Stress Summary	10	10	28	28

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	247	361	380	434

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1205	1355	1205	1355	1205	1355	1205	1355
Circ. Pl (SUS)	-7	-7	29	29	-9	-9	31	31
Circ. Q (SUS)	-21	21	91	-91	-140	140	352	-352
Long. Pm (SUS)	602	602	602	602	602	602	602	602
Long. Pl (SUS)	-3	-3	25	25	5	5	17	17
Long. Q (SUS)	-49	49	261	-261	-55	55	125	-125
Shear Pm (SUS)	0	0	0	0	0	0	0	0



Shear P1 (SUS)	12	12	-12	-12	-9	-9	9	9
Shear Q (SUS)	21	21	21	21	21	21	21	21
Pm (SUS)	1205	1355	1205	1355	1205	1355	1205	1355
Pm+P1 (SUS)	1198	1348	1234	1384	1196	1346	1236	1386
Pm+P1+Q (Total)	1178	1370	1325	1293	1056	1486	1589	1035

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1355.00	1407.13	Passed
Pm+P1 (SUS)	1386.11	2110.69	Passed
Pm+P1+Q (TOTAL)	1589.06	4221.39	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	175	355	175	355	175	355	175	355
Circ. P1 (SUS)	-7	-7	29	29	5	5	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	87	87	87	87	87	87	87	87
Long. P1 (SUS)	-57	-57	71	71	-57	-57	71	71
Long. Q (SUS)	-33	33	153	-153	-96	96	216	-216
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	11	11	-11	-11	-8	-8	8	8
Shear Q (SUS)	19	19	19	19	19	19	19	19
Pm (SUS)	175.0	355.0	175.0	355.0	175.0	355.0	175.0	355.0
Pm+P1 (SUS)	168.9	348.4	206.5	384.5	180.4	360.2	193.8	372.3
Pm+P1+Q (Total)	181.2	351.1	311.6	384.2	247.0	360.5	377.9	433.4

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	355.00	1407.13	Passed
Pm+P1 (SUS)	384.53	2110.70	Passed
Pm+P1+Q (TOTAL)	433.38	4221.40	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Input, Nozzle Desc: T1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	150.053	kgf/cm ²
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1510.00	mm
Shell Finished (Minimum) Thickness	t	92.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		137.82	cm
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

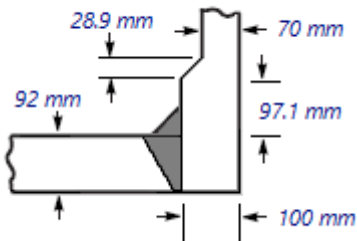
Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	70.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	97.1325	mm
Height of Beveled Transition	L'	28.8675	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	90.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T1-Tube Inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	10.000	in.
Actual Thickness Used in Calculation	2.756	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 260 \cdot (\exp(150/(1407 \cdot 1)) - 1) + 3 + 0$
 $= 14.6 + 3 + 0 = 17.6 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 67 + 260) / 260) - 0$
 $= 585 - 0 = 585 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 88.384 mm

Nozzle neck minimum required thickness : 11.340 mm

Minimum nozzle neck thickness 70.002 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 257.9,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3259.9 kgf/cm ²	Passed
Occasional	: 115.7,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 123.2,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1407/1407, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 100 - 3$
 $= 97.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 70 - 3$
 $= 67.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1516/89$
 $= 17.034 \text{ must be less than } 400.$



Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1510/2 + 3$$

$$= 758.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{758 \cdot 89}, 2 \cdot 130)$$

$$= 259.734 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{\frac{R_n \cdot t_n}{t}}$$

$$= \min[1.5 \cdot 89, 0] + \sqrt{130 \cdot 97}$$

$$= 112.294 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(89 + 0)$$

$$= 712.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[112, 300, 712] + 89$$

$$= 201.294 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot frp) / LR)$$

$$= 89 + ((0 \cdot 1) / 260)$$

$$= 89.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 130 + 97) / (\sqrt{(1516 + 89) \cdot 89}))$$

$$= 0.945$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 89 \cdot 260 \cdot \max((0.945/5)^{0.85}, 1)$$

$$= 231.164 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2B}$$

$$= 181 + 13.5$$

$$= 194.061 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + \text{frn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 231 + 1(194 + 0) + 0.5 + 0 + 0$$

$$= 425.725 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 97 / \ln[1 + 97/130]$$

$$= 174.017 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 89 / \ln[1 + 89/758]$$

$$= 801.677 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407 \cdot 1$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 150 \cdot 174 \cdot 201$$

$$= 52561.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs (LR + tn)$$

$$= 150 \cdot 802 (260 + 97)$$

$$= 429128.9 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 150 \cdot 802 \cdot 130$$

$$= 156381.7 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (429129 + 156382 + 52561) / 150$$

$$= 4252.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2111 / (2 \cdot 4252 / 426 - 802 / 89)$$

$$= 192.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 1407 [89/802]$$

$$= 156.2 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(192, 156)$$

$$= 156.216 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (52561 + 429129 + 156382) / 426$$

$$= 1498.793 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 150 \cdot 802 / 89$$

$$= 1351.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 1499 - 1352, 1352)$$

$$= 1646.0 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1645.97	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	156.22	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (130 + 97) / 130$$

$$= 1.746 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (130 + 97)$$

$$= 356.571 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 97.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(156382 \cdot 1.75, 1.5 \cdot 1407(194+0), \pi/4 \cdot 150 \cdot 130^2 \cdot 1.75^2)$$

$$= 60727.438 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 60727 / (357(0.49 \cdot 7.07 + 0.6 \cdot 89 + 0.49 \cdot 0))$$

$$= 299.500 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle away from the intersecting weld by 188.258 mm.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 0 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.218

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (14.6 \cdot 1) / (70 - 3 - 0)$$

$$= 0.218$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 18 °C
User entered Minimum Design Temperature	: 0 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: 20 °C
Temperature reduction per Graph 3.13	: 2 °C
The computed stress ratio is	: 0.958

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (85.3 \cdot 1) / (92 - 3 - 0)$$

$$= 0.958$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 18 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -42 °C

The Drop for this Nozzle is : 34.9333 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.9333 mm

Input Echo, WRC297 Item 1, Description: T1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1510.000	mm
Shell Thickness	92.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	254.000	mm
Nozzle Thickness	100.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	150.05	kgf/cm ²
Radial Load	P	-8158.12	kgf
Circumferential Shear	Vc	8158.12	kgf
Longitudinal Shear	Vl	6118.59	kgf
Circumferential Moment	Mc	5300.64	kgf-m.
Longitudinal Moment	Ml	5300.64	kgf-m.
Torsional Moment	Mt	6116.12	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (18.413) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tv, (4.935) is < 5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (4.540) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 454.0 + 2 \cdot 1.65 \cdot \sqrt{(802.5(92.0 - 3.0))}$
 $= 1335.925 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1605.000	mm
Nozzle Outside Diameter	(d) =	454.000	mm
Vessel Thickness used	(T) =	89.000	mm
Nozzle Thickness used	(t) =	97.000	mm
T / t	=	0.918	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.201$$

Nr/P	=	0.112
Mr/P	=	0.172
M0/P	=	0.057
N0/P	=	0.116
MrD/Mc	=	0.279
NrDL/Mc	=	0.137
M0d/Mc	=	0.102
N0DL/Mc	=	0.044



MrD/M1 = 0.176
 NrDL/M1 = 0.100
 MØD/M1 = 0.064
 NØDL/M1 = 0.128

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	11	11	11	11
Outplane Bending (P)	35	-35	35	-35
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-56	56	56	-56
Normal Pressure Stress	1207	1357	1207	1357
Outplane Stress Summary	1178	1371	1329	1296

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	106	-106	106	-106
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-14	-14	14	14
Inplane Bending (ML)	-155	155	155	-155
Inplane Pressure Stress	603	603	603	603
Inplane Stress Summary	551	649	891	367

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	21	21	21	21
Shear Stress Summary	33	33	8	8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1180	1372	1329	1296

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	11	11	11	11
Outplane Bending (P)	35	-35	35	-35
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-90	90	90	-90
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	603	603	603	603
Outplane Stress Summary	554	663	747	496

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	106	-106	106	-106
Inplane Membrane (Mc)	-20	-20	20	20
Inplane Bending (Mc)	-246	246	246	-246
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1207	1357	1207	1357
Inplane Stress Summary	1058	1488	1591	1035

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	21	21	21	21
Shear Stress Summary	11	11	30	30

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1058	1488	1592	1037

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	11	11	11	11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	175	355	175	355
Hoop Stress Summary	168	348	206	386

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	7	7	7	7
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-64	-64	64	64
Axial Bending (ML)	-93	93	93	-93
Axial Pressure Stress	87	87	87	87
Axial Stress Summary	-2	63	313	6

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	11	11	-11	-11
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	19	19	19	19
Shear Stress Summary	31	31	7	7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	182	352	314	386

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	11	11	11	11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	175	355	175	355
Hoop Stress Summary	181	361	193	374

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	7	7	7	7
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	-64	-64	64	64
Axial Bending (Mc)	-156	156	156	-156
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	87	87	87	87
Axial Stress Summary	-65	126	376	-56

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-8	-8	8	8
Shear due to Torsion	19	19	19	19
Shear Stress Summary	10	10	28	28

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	247	361	380	434

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:
Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1207	1357	1207	1357	1207	1357	1207	1357
Circ. Pl (SUS)	-7	-7	29	29	-9	-9	31	31
Circ. Q (SUS)	-21	21	91	-91	-140	140	352	-352
Long. Pm (SUS)	603	603	603	603	603	603	603	603
Long. Pl (SUS)	-3	-3	25	25	5	5	17	17
Long. Q (SUS)	-49	49	261	-261	-55	55	125	-125
Shear Pm (SUS)	0	0	0	0	0	0	0	0



Shear P1 (SUS)	12	12	-12	-12	-9	-9	9	9
Shear Q (SUS)	21	21	21	21	21	21	21	21
Pm (SUS)	1207	1357	1207	1357	1207	1357	1207	1357
Pm+P1 (SUS)	1200	1350	1236	1386	1198	1348	1238	1388
Pm+P1+Q (Total)	1180	1372	1327	1295	1058	1488	1591	1037

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1357.00	1407.13	Passed
Pm+P1 (SUS)	1388.11	2110.69	Passed
Pm+P1+Q (TOTAL)	1591.06	4221.39	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	175	355	175	355	175	355	175	355
Circ. P1 (SUS)	-7	-7	29	29	5	5	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	87	87	87	87	87	87	87	87
Long. P1 (SUS)	-57	-57	71	71	-57	-57	71	71
Long. Q (SUS)	-33	33	153	-153	-96	96	216	-216
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	11	11	-11	-11	-8	-8	8	8
Shear Q (SUS)	19	19	19	19	19	19	19	19
Pm (SUS)	175.0	355.0	175.0	355.0	175.0	355.0	175.0	355.0
Pm+P1 (SUS)	168.9	348.4	206.5	384.5	180.4	360.2	193.8	372.3
Pm+P1+Q (Total)	181.2	351.1	311.6	384.2	247.0	360.5	377.9	433.4

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	355.00	1407.13	Passed
Pm+P1 (SUS)	384.53	2110.70	Passed
Pm+P1+Q (TOTAL)	433.38	4221.40	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: S1-Shell Inlet From: 40

Pressure for Reinforcement Calculations	P	194.770	kgf/cm ²
Temperature for Internal Pressure	Temp	93	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1510.00	mm
Shell Finished (Minimum) Thickness	t	118.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		345.82	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

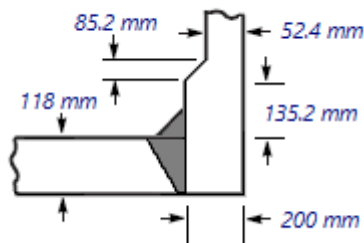
Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		140	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	135.2310	mm
Height of Beveled Transition	L'	85.2314	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	118.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: S1-Shell Inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	19.876	in.
Actual Thickness Used in Calculation	2.062	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 511 \cdot (\exp(195/(1407 \cdot 1)) - 1) + 3 + 0$
 $= 37.9 + 3 + 0 = 40.9 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 49.4 + 511) / 511) - 0$
 $= 249 - 0 = 249 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 115.721 mm

Nozzle neck minimum required thickness : 11.340 mm

Minimum nozzle neck thickness 45.822 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 510.0,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3007.8 kgf/cm ²	Passed
Occasional	: 535.9,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 71.4,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1407/1407, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 200 - 3$
 $= 197.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 52.4 - 3$
 $= 49.375 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1516/115$
 $= 13.183 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:

$$= Di/2 + \text{corrosion allowance}$$

$$= 1510/2 + 3$$

$$= 758.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$$

$$= \min(\sqrt{758 \cdot 115}, 2 \cdot 255)$$

$$= 295.246 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, te] + \sqrt{Rn \cdot tn}$$

$$= \min[1.5 \cdot 115, 0] + \sqrt{255 \cdot 197}$$

$$= 224.318 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= Lpr1$$

$$= 300$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + te)$$

$$= 8(115 + 0)$$

$$= 920.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[224, 300, 920] + 115$$

$$= 339.318 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 115 + ((0 \cdot 1) / 295)$$

$$= 115.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 255 + 197) / (\sqrt{(1516 + 115) \cdot 115}))$$

$$= 1.634$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 115 \cdot 295 \cdot \max((1.63/5)^{0.85}, 1)$$

$$= 339.533 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 493 + 43.2$$

$$= 536.205 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg41^2$$

$$= 0.5 \cdot 10^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 340 + 1(536 + 0) + 0.5 + 0 + 0$$

$$= 876.237 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 197 / \ln[1 + 197/255]$$

$$= 344.591 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 115 / \ln[1 + 115/758]$$

$$= 814.147 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1407 \cdot 1$$

$$= 2110.7 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 195 \cdot 345 \cdot 339$$

$$= 227736.8 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 195 \cdot 814 (295 + 197)$$

$$= 780561.2 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 195 \cdot 814 \cdot 255$$

$$= 405031.5 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (780561 + 405032 + 227737) / 195$$

$$= 7256.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2111 / (2 \cdot 7256 / 876 - 814 / 115)$$

$$= 222.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 1407 [115 / 814]$$

$$= 198.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(223, 199)$$

$$= 198.760 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (227737 + 780561 + 405032) / 876$$

$$= 1612.955 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 195 \cdot 814 / 115$$

$$= 1378.9 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1613 - 1379, 1379)$$

$$= 1847.0 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1847.03	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	198.76	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (255 + 197) / 255$$

$$= 1.771 \text{ For set-in Nozzles}$$



Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (255 + 197)$$

$$= 710.668 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 197.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(405032 \cdot 1.77, 1.5 \cdot 1407(536+0), \pi/4 \cdot 195 \cdot 255^2 \cdot 1.77^2)$$

$$= 313116.406 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 313116 / (711(0.49 \cdot 7.07 + 0.6 \cdot 115 + 0.49 \cdot 0))$$

$$= 608.013 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	23 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	28 °C
Temperature reduction per Graph 3.13	:	6 °C
The computed stress ratio is	:	0.885

The computed thickness ratio [sr]:

$$= (t_r \cdot E_s) / (t_{nom} - c_i - c_{ext})$$

$$= (37.9 \cdot 1) / (45.8 - 3 - 0)$$

$$= 0.885$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 23 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -32 °C

The Drop for this Nozzle is : 150.5694 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 568.5693 mm

Input Echo, WRC297 Item 1, Description: S1-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1510.000	mm
Shell Thickness	118.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm



Nozzle Diameter	504.850	mm
Nozzle Thickness	200.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	1407.13 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.13 kgf/cm ²

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	194.77	kgf/cm ²
Radial Load	P	-19579.53	kgf
Circumferential Shear	Vc	19579.53	kgf
Longitudinal Shear	Vl	14684.63	kgf
Circumferential Moment	Mc	30.53	kgf-m.
Longitudinal Moment	Ml	30.53	kgf-m.
Torsional Moment	Mt	35.23	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (14.797) is < 20
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Dv (.518) is > 0.5
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (4.524) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 904.85 + 2 \cdot 1.65 \cdot \sqrt{(815.5(118.0 - 3.0))}$
 $= 1915.440 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1631.000 mm
Nozzle Outside Diameter	(d) =	904.850 mm
Vessel Thickness used	(T) =	115.000 mm
Nozzle Thickness used	(t) =	197.000 mm
T / t	=	0.584
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.089$$

Nr/P	=	0.070
Mr/P	=	0.134
M0/P	=	0.040
N0/P	=	0.097
MrD/Mc	=	0.276
NrDL/Mc	=	0.108
M0d/Mc	=	0.087
N0DL/Mc	=	0.069
MrD/Ml	=	0.110
NrDL/Ml	=	0.064
M0D/Ml	=	0.036
N0DL/Ml	=	0.122

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	35	-35	35	-35
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1193	1388	1193	1388
Outplane Stress Summary	1242	1366	1242	1366

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	118	-118	118	-118
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	596	596	596	596
Inplane Stress Summary	725	488	726	487

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	11	11	-11	-11
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	11	11	-11	-11

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1243	1367	1243	1367

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	35	-35	35	-35
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	596	596	596	596
Outplane Stress Summary	646	575	646	575

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	118	-118	118	-118
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1193	1388	1193	1388
Inplane Stress Summary	1322	1279	1323	1279

Vessel Stresses (kgf/cm²):



CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-8	-8	8	8

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1322	1279	1323	1279

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	14	14	14	14
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	218	452	218	452
Hoop Stress Summary	232	466	232	466

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	29	-29	29	-29
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	109	109	109	109
Axial Stress Summary	143	83	143	83

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	6	6	-6	-6
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	6	6	-6	-6

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	233	466	233	466

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	14	14	14	14
Hoop Bending (P)	0	0	0	0

Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	218	452	218	452

Hoop Stress Summary	232	466	232	466
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	4	4	4	4
Axial Bending (P)	29	-29	29	-29
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	109	109	109	109

Axial Stress Summary	143	83	143	83
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-5	-5	5	5
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	233	466	233	466
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1193	1388	1193	1388	1193	1388	1193	1388
Circ. Pl (SUS)	14	14	14	14	10	10	10	10
Circ. Q (SUS)	35	-35	35	-35	118	-118	118	-118
Long. Pm (SUS)	596	596	596	596	596	596	596	596
Long. Pl (SUS)	10	10	10	10	14	14	14	14
Long. Q (SUS)	118	-118	118	-118	35	-35	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-8	-8	8	8
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1193	1388	1193	1388	1193	1388	1193	1388
Pm+Pl (SUS)	1207	1402	1207	1402	1203	1398	1203	1398
Pm+Pl+Q (Total)	1242	1367	1242	1367	1321	1280	1321	1280

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	1388.00	1407.13	Passed
Pm+Pl (SUS)	1402.15	2110.69	Passed
Pm+Pl+Q (TOTAL)	1367.14	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	218	452	218	452	218	452	218	452
Circ. Pl (SUS)	14	14	14	14	14	14	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	109	109	109	109	109	109	109	109
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	29	-29	29	-29	29	-29	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	6	6	-6	-6	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	218.0	452.0	218.0	452.0	218.0	452.0	218.0	452.0
Pm+Pl (SUS)	232.3	466.1	232.3	466.1	232.2	466.1	232.2	466.1
Pm+Pl+Q (Total)	232.4	466.1	232.4	466.1	232.3	466.1	232.3	466.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	452.00	1407.13	Passed
Pm+Pl (SUS)	466.10	2110.70	Passed
Pm+Pl+Q (TOTAL)	466.09	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: S2-Shell outlet

From: 40

Pressure for Reinforcement Calculations	P	194.778	kgf/cm ²
Temperature for Internal Pressure	Temp	93	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1510.00	mm
Shell Finished (Minimum) Thickness	t	118.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		695.82	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

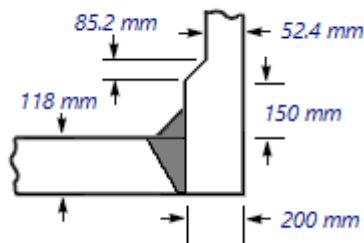
Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		140	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	85.2314	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	118.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: S2-Shell outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	19.876	in.
Actual Thickness Used in Calculation	2.062	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 511 \cdot (\exp(195/(1407 \cdot 1)) - 1) + 3 + 0$$

$$= 37.9 + 3 + 0 = 40.9 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1407 \cdot 1 \cdot \ln((2 \cdot 49.4 + 511) / 511) - 0$$

$$= 249 - 0 = 249 \text{ kgf/cm}^2$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 511.87 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 747.67 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 115.726 mm

Nozzle neck minimum required thickness : 11.340 mm

Minimum nozzle neck thickness 45.822 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 510.1,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3007.8 kgf/cm ²	Passed
Occasional	: 535.9,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 71.4,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1407/1407, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 200 - 3$$

$$= 197.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 52.4 - 3$$

$$= 49.375 \text{ mm}$$



Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1516/115$$

$$= 13.183 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1510/2 + 3$$

$$= 758.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{758 \cdot 115}, 2 \cdot 255)$$

$$= 295.246 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 115, 0] + \sqrt{255 \cdot 197}$$

$$= 224.318 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(115 + 0)$$

$$= 920.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[224, 300, 920] + 115$$

$$= 339.318 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 115 + ((0 \cdot 1) / 295)$$

$$= 115.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 255 + 197) / (\sqrt{(1516 + 115) \cdot 115}))$$

$$= 1.634$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 115 \cdot 295 \cdot \max((1.63/5)^{0.85}, 1)$$

$$= 339.533 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 522 + 98.6$$

$$= 620.625 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \min(A2 + A3) + A41 + A42 + A43$$

$$= 340 + 1(621 + 0) + 0.5 + 0 + 0$$

$$= 960.657 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 197 / \ln[1 + 197/255]$$

= 344.591 mm

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 = $115 / \ln[1 + 115/758]$
 = 814.147 mm

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S \cdot E$
 = $1.5 \cdot 1407 \cdot 1$
 = 2110.7 kgf/cm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
 = $195 \cdot 345 \cdot 339$
 = 227745.7 kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
 = $195 \cdot 814 (295 + 197)$
 = 780591.6 kgf

Discontinuity Force from Internal Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
 = $195 \cdot 814 \cdot 255$
 = 405047.3 kgf

Area Resisting Internal Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
 = $(780592 + 405047 + 227746) / 195$
 = 7256.4 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
 = $2111 / (2 \cdot 7256 / 961 - 814 / 115)$
 = 262.9 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S / [t / R_{xs}]$
 = $1407 [115 / 814]$
 = 198.8 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
 = $\min(263, 199)$
 = 198.760 kgf/cm²

Average Primary Membrane Stress [σ_{Avg}]:

= $(f_N + f_S + f_Y) / A_T$
 = $(227746 + 780592 + 405047) / 961$
 = 1471.270 kgf/cm²

General Primary Membrane Stress [σ_{Circ}]:

= $P \cdot R_{xs} / t_{eff}$
 = $195 \cdot 814 / 115$
 = 1378.9 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 = $\max(2 \cdot 1471 - 1379, 1379)$
 = 1563.6 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1563.60	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	198.76	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:



$$= (R_{nc} + t_n) / R_{nc}$$

$$= (255 + 197) / 255$$

$$= 1.771 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (255 + 197)$$

$$= 710.668 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 197.000, 6.000) \text{ Passed}$$

Weld Load Value [f_{welds}]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(405047 \cdot 1.77, 1.5 \cdot 1407(621+0), \pi/4 \cdot 195 \cdot 255^2 \cdot 1.77^2)$$

$$= 313128.531 \text{ kgf}$$

Weld Stress Value [τ]:

$$= f_{welds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 313129 / (711(0.49 \cdot 7.07 + 0.6 \cdot 115 + 0.49 \cdot 0))$$

$$= 608.036 < \text{ or } = \text{ to } 1407.129 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 23 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 28 °C
Temperature reduction per Graph 3.13	: 6 °C
The computed stress ratio is	: 0.885

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (37.9 \cdot 1) / (45.8 - 3 - 0)$$

$$= 0.885$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 23 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -32 °C

The Drop for this Nozzle is : 150.5694 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 568.5693 mm

Input Echo, WRC297 Item 1, Description: S2-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1510.000	mm
Shell Thickness	118.0000	mm
Shell Stress Concentration Factor	1.200	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²



Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		504.850	mm
Nozzle Thickness		200.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.13	kgf/cm ²

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	194.78	kgf/cm ²
Radial Load	P	-19579.53	kgf
Circumferential Shear	Vc	19579.53	kgf
Longitudinal Shear	Vl	14684.63	kgf
Circumferential Moment	Mc	30.53	kgf-m.
Longitudinal Moment	Ml	30.53	kgf-m.
Torsional Moment	Mt	35.23	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

*Warning - The ratio, Dv/Tv (14.797) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Dv (.518) is > 0.5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (4.524) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 904.85 + 2 \cdot 1.65 \cdot \sqrt{(815.5(118.0 - 3.0))}$
 $= 1915.440 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1631.000	mm
Nozzle Outside Diameter	(d) =	904.850	mm
Vessel Thickness used	(T) =	115.000	mm
Nozzle Thickness used	(t) =	197.000	mm
T / t	=	0.584	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.089$$

Nr/P	=	0.070
Mr/P	=	0.134
M0/P	=	0.040
N0/P	=	0.097
MrD/Mc	=	0.276
NrDL/Mc	=	0.108
M0d/Mc	=	0.087
N0DL/Mc	=	0.069
MrD/Ml	=	0.110
NrDL/Ml	=	0.064
M0D/Ml	=	0.036
N0DL/Ml	=	0.122

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	42	-42	42	-42
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1193	1388	1193	1388

Outplane Stress Summary	1252	1362	1252	1362
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	142	-142	142	-142
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	596	596	596	596

Inplane Stress Summary	751	466	752	466
------------------------	-----	-----	-----	-----

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	11	11	-11	-11
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	11	11	-11	-11
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1253	1362	1253	1362
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	42	-42	42	-42
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	596	596	596	596

Outplane Stress Summary	656	571	656	571
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	142	-142	142	-142
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1193	1388	1193	1388



Inplane Stress Summary 1347 1258 1348 1257

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary -8 -8 8 8

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress 1348 1258 1349 1257

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Stresses in the
 hoop direction)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	14	14	14	14
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	218	452	218	452

Hoop Stress Summary 232 466 232 466

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	4	4	4	4
Axial Bending (P)	29	-29	29	-29
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	109	109	109	109

Axial Stress Summary 143 83 143 83

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	6	6	-6	-6
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 6 6 -6 -6

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress 233 466 233 466

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses in the

	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	14	14	14	14
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	218	452	218	452
Hoop Stress Summary	232	466	232	466

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	29	-29	29	-29
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	109	109	109	109
Axial Stress Summary	143	83	143	83

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	233	466	233	466

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1193	1388	1193	1388	1193	1388	1193	1388
Circ. Pl (SUS)	17	17	17	17	12	12	12	12
Circ. Q (SUS)	42	-42	42	-42	142	-142	142	-142
Long. Pm (SUS)	596	596	596	596	596	596	596	596
Long. Pl (SUS)	12	12	12	12	17	17	17	17
Long. Q (SUS)	142	-142	142	-142	42	-42	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-8	-8	8	8
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1193	1388	1193	1388	1193	1388	1193	1388
Pm+Pl (SUS)	1210	1405	1210	1405	1205	1400	1205	1400
Pm+Pl+Q (Total)	1252	1363	1252	1363	1347	1258	1347	1258

Vessel Stress Summation Comparison (kgf/cm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1388.00	1407.13	Passed
Pm+Pl (SUS)	1405.15	2110.69	Passed
Pm+Pl+Q (TOTAL)	1363.13	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	218	452	218	452	218	452	218	452
Circ. Pl (SUS)	14	14	14	14	14	14	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	109	109	109	109	109	109	109	109
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	29	-29	29	-29	29	-29	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	6	6	-6	-6	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	218.0	452.0	218.0	452.0	218.0	452.0	218.0	452.0
Pm+Pl (SUS)	232.3	466.1	232.3	466.1	232.2	466.1	232.2	466.1
Pm+Pl+Q (Total)	232.4	466.1	232.4	466.1	232.3	466.1	232.3	466.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	452.00	1407.13	Passed
Pm+Pl (SUS)	466.10	2110.70	Passed
Pm+Pl+Q (TOTAL)	466.09	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	194.77	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	194.77	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00	kgf/cm ²
Shell Thickness	ts	118.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1510.000	mm
Shell Circumferential Joint Efficiency	Esw	1.000	
Shell Temperature for Internal Pressure	Ts	200.00	°C
Shell Material		SA-516 70	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700	
Shell Allowable Stress at Temperature	Ss	1529.51	kgf/cm ²
Shell Allowable Stress at Ambient		1784.41	kgf/cm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	149.90	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	149.90	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	92.0000	mm
Channel Internal Corrosion Allowance	cac	3.0000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1510.000	mm
Channel Design Temperature	TEMPC	85.00	°C
Channel Flange Matl		SA-516 70	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700	
Channel Allowable Stress at Temperature	Sc	1643.13	kgf/cm ²
Channel Allowable Stress at Ambient		1784.41	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	1548	
Tube Wall Thickness	et	3.0480	mm
Tube Outside Diameter	D	19.0500	mm
Total Straight Tube Length	Lt	7315.00	mm
Straight Tube Length (bet. inner tubsht faces) L		6755.00	mm
Design Temperature of the Tubes		200.00	°C
Tube Material		SA-179	
Is this a Welded Tube		No	
Tube Allowable Stress at Temperature		942.17	kgf/cm ²
Tube Allowable Stress At Ambient		942.17	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1570.29	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	30.4500	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.4130	mm
Groove Weld Leg	ag	2.4130	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		4.21.2	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	719.161	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	0.0000	mm
Area of Tube Layout	Ap	12174.7	cm ²
Perimeter of Tube Layout	Cp	4316.4	mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 200.00 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 1407.16 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.16 kgf/cm²
 Thickness of Tubesheet h 280.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 6.0000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 6.0000 mm
 Tubesheet Outside Diameter A 1746.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

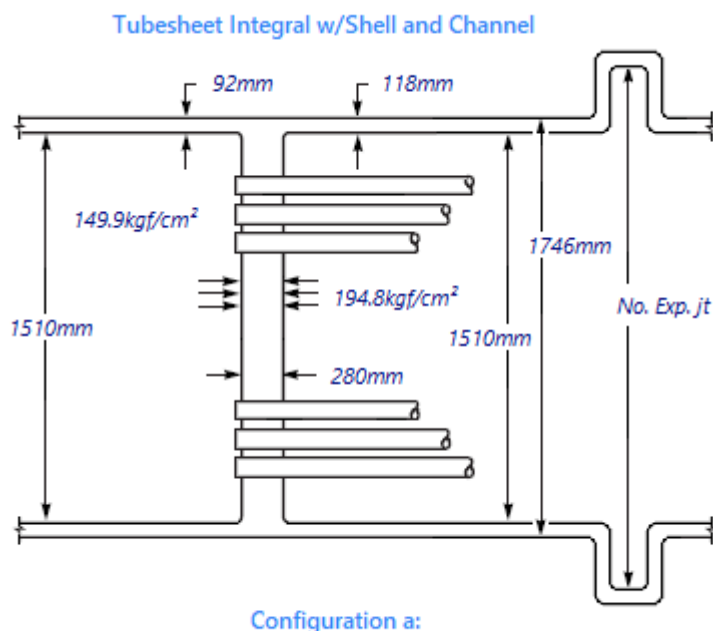
Unsupported Tube Span under consideration l 937.000 mm
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 41.82 °C
 Mean Tube Metal Temp. along Tube length Ttm 40.28 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%



Th. Exp. Coeff. Metal Temp. along Len	41.8 °C	0.0000117252 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19638E+07 kgf/cm²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	41.8 °C	0.20549E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.20297E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	40.3 °C	0.0000117152 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19638E+07 kgf/cm²
Elastic Mod. at Metal Temp. along Len	40.3 °C	0.20558E+07 kgf/cm²
Elastic Mod. at Tubsht. Design Temp.	200.0 °C	0.19638E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	200.0 °C	0.19638E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 13 \cdot (\exp(150/(942 \cdot 1)) - 1) + 0$
 $= 1.12 + 0$
 $= 1.12 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 6755 / \sqrt{(9.52 \cdot 3.05)}$
 $= 1253.676$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19/3.05)$
 $= 0.0880$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y(t / Do)$
 $= 1.6 \cdot 0.088 \cdot 1963824 (3.05/19)$
 $= 44241.027 \text{ kgf/cm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 44241.0/1570.3$
 $= 28.174$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 1570/3304$
 $= 0.475$

Ratio [AE]:

$$= F_{he} / E$$

$$= 44241/1963824$$

$$= 0.022528$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.48) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$$

$$= 0.162$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 1570 (1 + 0.0020) / \ln(1 + 0.0020) 0.16202243$$

$$= 4307.376 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6 (1 - 0.48)$$

$$= 0.315$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (3304 \cdot \exp(0.31)) / (0.31 0.31)$$

$$= 6514.311 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.481.5 - 0.5 \cdot 0.482.5 - 0.483.5$$

$$= 0.340$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	1652	-1.722	0.505E-05	0.418E-05	0.123E-04	-0.894E-05	0.763E+05	0.864E-03
2	2478	1.084	0.411E-04	-0.533E-04	0.298E-04	0.527E-04	0.141E+05	0.153E+00
3	2065	-0.319	0.160E-04	-0.115E-04	0.200E-04	0.338E-04	0.170E+05	0.990E-01
4	1858	-1.020	0.928E-05	0.328E-05	0.159E-04	0.625E-06	0.338E+05	0.325E-01
5	1755	-1.371	0.690E-05	0.455E-05	0.141E-04	-0.635E-05	0.508E+05	0.120E-01
6	1703	-1.546	0.592E-05	0.448E-05	0.132E-04	-0.804E-05	0.624E+05	0.479E-02
7	1678	-1.634	0.547E-05	0.435E-05	0.127E-04	-0.858E-05	0.690E+05	0.179E-02
8	1665	-1.678	0.525E-05	0.427E-05	0.125E-04	-0.878E-05	0.725E+05	0.424E-03
9	1658	-1.700	0.515E-05	0.423E-05	0.124E-04	-0.887E-05	0.744E+05	0.230E-03
10	1661	-1.689	0.520E-05	0.425E-05	0.125E-04	-0.882E-05	0.735E+05	0.949E-04
11	1660	-1.694	0.517E-05	0.424E-05	0.125E-04	-0.884E-05	0.739E+05	0.680E-04
12	1661	-1.692	0.519E-05	0.424E-05	0.125E-04	-0.883E-05	0.737E+05	0.133E-04
13	1660	-1.693	0.518E-05	0.424E-05	0.125E-04	-0.884E-05	0.738E+05	0.274E-04
14	1660	-1.692	0.518E-05	0.424E-05	0.125E-04	-0.884E-05	0.737E+05	0.705E-05
15	1660	-1.692	0.519E-05	0.424E-05	0.125E-04	-0.884E-05	0.737E+05	0.312E-05
16	1660	-1.692	0.518E-05	0.424E-05	0.125E-04	-0.884E-05	0.737E+05	0.197E-05

Factor of Safety [FS]:

$$\text{Since } F_{ic} [1661] \geq F_y [1570], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 1661 / 1.67$$

$$= 996.363 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 996 (3.05 / 19)$$

$$= 318.836 \text{ kgf/cm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.1123 mm
Allowable Internal Pressure at Corroded thickness	363.36 kgf/cm ²
Required Internal Design Pressure	149.90 kgf/cm ²
Allowable External Pressure at Corroded thickness	318.84 kgf/cm ²

Required External Design Pressure 194.77 kgf/cm²
 Required Thickness due to Shell Side pressure 2.1123 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:
 $= l_{tx} / h$ (modified for corrosion if present)
 $= 0/280$
 $= 0.0000$ (must be $0 \leq p \leq 1$)

Effective tube hole diameter [d*]:
 $= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (p); dt - 2 \cdot tt)$
 $= \max(19 - 2 \cdot 3.05 (1963824/1963824) \cdot (0); 19 - 2 \cdot 3.05)$
 $= 19.0500$ mm

Equivalent outer tube limit circle diameter [Do]:
 $= 2 \cdot ro + dt$
 $= 2 \cdot 719 + 19$
 $= 1457.372$ mm

Determine the basic ligament efficiency for shear [μ]:
 $= (p - dt) / p$
 $= (30.5 - 19) / 30.5$
 $= 0.374$

Equivalent outer tube limit radius [ao]:
 $= Do / 2$
 $= 1457/2$
 $= 728.6860$ mm

Effective tube pitch [p*]:
 $= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$
 $= 30.5 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1457 \cdot 30.5) / (\pi \cdot 1457^2)) }$
 $= 30.4500$ mm

Effective ligament efficiency for bending [μ*]:
 $= (p* - d*) / p*$
 $= (30.5 - 19) / 30.5$
 $= 0.3744$

Ratio [ps]:
 $= as / ao$
 $= 755/729$
 $= 1.0361$

Ratio [pc]:
 $= ac / ao$
 $= 755/729$
 $= 1.0361$

Compute parameter [xt]:
 $= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$
 $= 1 - 1548 ((19 - 2 \cdot 3.05) / (2 \cdot 729))^2$
 $= 0.8777$

Determine parameter [xs]:
 $= 1 - Nt(dt / (2 \cdot ao))^2$
 $= 1 - 1548 (19 / (2 \cdot 729))^2$
 $= 0.7355$

Determine the value [h'g]:
 $= \max((h_g - CATC); 0)$ (For pressure only cases)
 $= \max((0 - 0); 0)$
 $= 0.000 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:
 $= \pi \cdot ts(D_s + ts)E_s/L$
 $= \pi \cdot 118 (1510 + 118)1963824/6755$
 $= 1754538.6250 \text{ kgf/mm}$

Axial Tube Stiffness [Kt]:
 $= \pi \cdot tt(D_t - tt)E_t/L$
 $= \pi \cdot 3.05 (19 - 3.05)1963824/6755$
 $= 445.4672 \text{ kgf/mm}$

Stiffness Factor [Ks,t]:
 $= K_s / (N_t \cdot K_t)$
 $= 1754538/(1548 \cdot 445)$
 $= 2.54435$

Rigidity Ratio [J]:
 $= 1 / (1 + K_s/K_t)$
 $= 1/(1 + 1754538/0)$
 $= 1.0000 (= 1 \text{ if No Exp. Jt.})$

Shell coefficient [β_s]:
 $= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts) ts)$
 $= ((12(1 - 0.3^2))^{1/4}) / ((1510 + 118) 118)$
 $= 0.0041 \text{ 1/mm}$

Shell coefficient [ks]:
 $= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$
 $= 0.0041 \cdot 1963824 \cdot 118^3 / (6(1 - 0.3^2))$
 $= 2451000576.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$

Shell coefficient [λ_s]:
 $= 6(D_s + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$
 $= 6(1510 + 118) \cdot 2.45E+09/280^3 \cdot (1 + 280 \cdot 0.0041 + (280 \cdot 0.0041)^2/2)$
 $= 3092588.5000 \text{ kgf/cm}^2$

Shell coefficient [δ_s]:
 $= D_s^2 / (4 \cdot E_s \cdot ts) \cdot (1 - D_s \cdot \nu_s / (2(D_s + ts)))$
 $= 1510^2 / (4 \cdot 1963824 \cdot 118) \cdot (1 - 1510 \cdot 0.3 / (2(1510 + 118)))$
 $= 0.21176E-02 \text{ mm/kgf/cm}^2$

Channel coefficient [β_c]:
 $= ((12(1 - \nu_c^2))^{1/4}) / ((D_c + tc) tc)$
 $= ((12(1 - 0.3^2))^{1/4}) / ((1510 + 92) 92)$
 $= 0.0047 \text{ 1/mm}$

Channel coefficient [kc]:
 $= \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_c^2))$
 $= 0.0047 \cdot 2029713 \cdot 92^3 / (6(1 - 0.3^2))$
 $= 1370680064.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:
 $CqDelta = CqTheta = CmTheta = CmDelta = 1$

Channel coefficient [λ_c]:
 $= 6(D_c + tc/\cos(\theta)) \cdot kc/h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot CqDelta)$
 $= 6(1510 + 92/\cos(0)) \cdot 1370680064/280^3 \cdot (1 + 280 \cdot 0.0047 \cdot (1 + 1)/2 + (280 \cdot 0.0047)^2/2 \cdot 1)$
 $= 1923399.4 \text{ kgf/cm}^2$

Channel coefficient [δ_c]:
 $= D_c^2 / (4 \cdot E_c \cdot Tc) \cdot (1 - D_c \cdot \nu_c / (2(D_c + tc)))$
 $= 1510^2 / (4 \cdot 2029713 \cdot 92) \cdot (1 - 1510 \cdot 0.3 / (2(1510 + 92)))$

$$= 0.26210\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:
 zetac = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 9.195403 ; \mu^* = 0.374384 ; E^* = 745274 \text{ kgf/cm}^2$$

$$E^*/E = 0.379501 ; \nu^* = 0.317620$$

Note: As h/p (9.195) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= \left(\left(24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.32^2)1548 \cdot 1963824 \cdot 3.05 \right. \right. \\ \left. \left. (19 - 3.05)729^2 \right) / (745274 \cdot 6755 \cdot 280^3) \right)^{1/4}$$

$$= 1.9801$$

Values from Table 4.18.3

$$Zd = 0.179335 ; Zv = 0.163904 ; Zm = 0.681118$$

$$Za = 0.785445\text{E}+00 ; Zw = 0.163904$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:
 = A / Do
 = $1746/1457$
 = 1.1980

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 745274 \cdot (3092588 + 1923399 + 1963824 \cdot \ln(1.2))$$

$$= 4.9176$$

Compute parameter [ϕ]:

$$= (1 + \nu^*)F$$

$$= (1 + 0.32)4.92$$

$$= 6.4795$$

Compute parameter [Q1]:

$$= (\psi_s - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm)$$

$$= (1.04 - 1 - 6.48 \cdot 0.16) / (1 + 6.48 \cdot 0.68)$$

$$= -0.189516$$

Compute Parameter [Qz1]:

$$= (Zd + Q1 \cdot Zw) / 2 \cdot Xa^4$$

$$= (0.18 + -0.19 \cdot 0.16) / 2 \cdot 1.98^4$$

$$= 1.1397$$

Compute parameter [Qz2]:

$$= (Zv + Q1 \cdot Zm) / 2 \cdot Xa^4$$

$$= (0.16 + -0.19 \cdot 0.68) / 2 \cdot 1.98^4$$

$$= 0.2677$$

Compute parameter [U]:

$$= (Zw + (\psi_s - 1)Zm)Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.16 + (1.04 - 1)0.68)1.98^4 / (1 + 6.48 \cdot 0.68)$$

$$= 0.5353$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:
 = 0 (config a)

Compute Parameter [γ]:
 = 0.000 mm (For Pressure only cases)

Calculate parameter [ω_s]:
 = $\psi_s \cdot k_s \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s)$

$$= 1.04 \cdot 2.45E+09 \cdot 0.0041 \cdot 0.00211762 (1 + 280 \cdot 0.0041)$$

$$= 48205.9883 \text{ mm}^2$$

Calculate parameter $[\omega*s]$:

$$= ao^2 (ps^2 - 1) \cdot (ps - 1) / 4 - \omega s$$

$$= 729^2 (1.04^2 - 1) \cdot (1.04 - 1) / 4 - 48206$$

$$= -47853.5273 \text{ mm}^2$$

Parameter $[\omega c]$:

$$= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.04 \cdot 1370680064 [0.002621014 (1 + 0.0047 \cdot 1 \cdot 280) -$$

$$0.000 (1/280 + 1 \cdot 0.0047/2)]$$

$$= 40994.0469 \text{ mm}^2$$

Parameter $[\omega*c]$:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega c$$

$$= 729^2 [(1.04^2 + 1) \cdot (1.04 - 1) / 4 - (1.04 - 1) / 2] - 40994$$

$$= -40641.5820 \text{ mm}^2$$

Pressure $[P*S]$:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure $[P*C]$:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure $[P's]$:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs -$$

$$[(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 195 \{ 0.74 + 2(1 - 0.74) 0.3 + [2/2.54 \cdot$$

$$(1510/1457)^2] 0.3 - [(1.04^2 - 1)/(1 \cdot 2.54)] -$$

$$[(1 - 1)/(2 \cdot 1 \cdot 2.54)] [(0^2 - (1510)^2)/1457^2] \}$$

$$= 217.8423 \text{ kgf/cm}^2$$

Pressure $[P't]$:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.88 + 2(1 - 0.88) 0.3 + 1/(1 \cdot 2.54)] 150$$

$$= 201.4843 \text{ kgf/cm}^2$$

Pressure $[P_Y]$:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 1548 \cdot 445 \cdot 0 / (3.14 \cdot 729^2)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Pressure $[P_w]$:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 0.54 \cdot 0 / (2 \cdot \pi \cdot 729^2)$$

$$= -0.0000 \text{ kgf/cm}^2$$

Calculate the pressure $[Prim]$:

$$= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= -(0.54/729^2) \cdot (-47854 \cdot 195 - -40642 \cdot 150)$$

$$= 3.2545 \text{ kgf/cm}^2$$

Calculate the pressure $[P\omega]$:

$$= (U/ao^2) \cdot (\omega s \cdot P*s - \omega c \cdot P*c)$$

$$= (0.54/729^2) \cdot (48206 \cdot 0 - 40994 \cdot 0)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Determine the effective pressure $[Pe]$:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot$$

$$(P's - P't + P_Y + P_w + P\omega + Prim)$$

$$= 0.1000E+01 \cdot 2.54 / (1 + 1 \cdot 2.54 (1.14 + (1.04 -$$

$$1) 0.27)) \cdot (218 - 201 + 0 + 0 + -0 + 3.25)$$

$$= 12.7157 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega_s \cdot P * s - \omega_c \cdot P * c)) + \\ W * \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\ = [((-47854 \cdot 195 - -40642 \cdot 150) - \\ (48206 \cdot 0 - 40994 \cdot 0)) \cdot 0.01 + \\ 0 \cdot 0 / (2 \cdot \pi)] / (1 + 6.48 \cdot 0.68) \\ = -5963.3975 \text{ kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot a o^2) \\ = -0.19 + 2 \cdot -5963 / (12.7 \cdot 729^2) \\ = -0.36616$$

Fm Value from Table 4.18.3 = 0.183081

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a o / (H - h'g))^2 \cdot Pe \\ = (1.5 \cdot 0.18 / 0.37) \cdot (2 \cdot 729 / (280 - 0))^2 \cdot 12.7 \\ = 252.6863 \text{ kgf/cm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 1407 = 2111 \text{ kgf/cm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a o / (h - h'g))^2 \cdot Pe \\ = (1.5 \cdot 0.024 / 0.37) \cdot (2 \cdot 729 / (16.7 - 0))^2 \cdot 2.91 \\ = 2110.6309 \text{ kgf/cm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc} \\ = 16.7 + 0 + 0 \\ = 16.6709 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a o) \cdot \min(0.8S; 0.533S_y) \\ = (2 \cdot 0.37 \cdot 280 / 729) \cdot 1126 \\ = 323.8907 \text{ kgf/cm}^2$$

Shear Stress is not required to be computed; [Pe] <= Tau_{limit}

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm} \\ \text{Tubesheet Shear Stress} = 201.8230 \text{ kgf/cm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS}) \\ = \max(16.7 ; 3.81) \\ = 16.6709 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.92, F_{tmax} = 1.03$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - Pe \cdot F_{tmin}) / (X_t - X_s) \\ = ((195 \cdot 0.74 - 150 \cdot 0.88) - (12.7) \cdot 0.92) / \\ (0.88 - 0.74) \\ = 0.1646 \text{ kgf/cm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - Pe \cdot F_{tmax}) / (X_t - X_s) \\ = ((195 \cdot 0.74 - 150 \cdot 0.88) - (12.7) \cdot 1.03) / \\ (0.88 - 0.74) \\ = -9.7116 \text{ kgf/cm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:



$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 9.71 \text{ kgf/cm}^2$$

Allowable Tube Stress $[\sigma_t, A]$:
 $= S_{ot} = 942.1682 \text{ kgf/cm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety $[F_s]$:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w))X_{a^4}); 1.25)$
 $= \max((3.25 - 0.25(0.18 + -0.37 \cdot 0.16))1.98^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2)$

Determine the factor $[r_t]$:
 $= (\sqrt{(dt^2 + (dt - 2 \cdot tt)^2)})/4$
 $= (\sqrt{(19^2 + (19 - 2 \cdot 3.05)^2)})/4$
 $= 5.7593 \text{ mm}$

Determine the factor $[C_t]$:
 $= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{yt})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 1963824 / 1570)}$
 $= 157.1179$

Determine the factor $[F_t]$:
 $= k \cdot L/r$
 $= 1 \cdot 937/5.76$
 $= 162.6940$

Buckling Allowable Stress $[S_{tb}]$:
 $= (1/F_s) \cdot (\pi^2 \cdot E_t / F_t^2) \text{ (} C_t \leq F_t \text{)}$
 $= (1/2) \cdot (\pi^2 \cdot 1963824 / 163^2)$
 $= 366.125 \text{ kgf/cm}^2 \text{ (Never greater than } S_{ot})$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load $[W_t]$:
 $= \sigma_{t,max} \cdot \text{Tube Area}$
 $= 9.71 \cdot 1.53$
 $= 14.88 \text{ kgf}$

Tube Metal Area $[A_t]$:
 $= \pi(d_o - t)t$
 $= \pi(19 - 3.05)3.05$
 $= 1.532 \text{ cm}^2$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength $[F_t]$:
 $= \pi \cdot t(d_o - t)S$
 $= \pi \cdot 3.05(19 - 3.05)942$
 $= 1443.667 \text{ kgf}$

Groove Weld Strength $[F_g]$:
 $= 0.85 \cdot \pi \cdot a_g(d_o + 0.67 \cdot a_g)S_w \text{ (but not } > F_t \text{)}$
 $= 0.85 \cdot \pi \cdot 2.41(19 + 0.67 \cdot 2.41)942$
 $= 1254.6581 \text{ kgf}$

Weld Strength Factor $[f_w]$:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 942/942$
 $= 1.000$

Design Strength Ratio $[f_d]$:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength $[F_f]$:
 $= 0.55 \cdot \pi \cdot a_f(d_o + 0.67 \cdot a_f)S_w \text{ (but not } > F_t \text{)}$
 $= 0.55 \cdot \pi \cdot 2.41(19 + 0.67 \cdot 2.41)942$
 $= 811.8376 \text{ kgf}$

Minimum Length of Weld $[a_r]$:

$$\begin{aligned}
 &= 2 \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd) - 0.75 \cdot do} \\
 &= 2 \sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 3.05 \cdot (19 - 3.05) \cdot 1 \cdot 1) - 0.75 \cdot 19} \\
 &= 3.445 \text{ mm}
 \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm
 Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 1444 is >= tube strength 1444 kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 1530 \cdot 1 = 1530 \text{ kgf/cm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 729^2 / ((1510 + 118) 118) \cdot [12.7 + (1.04^2 - 1)(195 - 150)] + 755^2 \cdot 150 / ((1510 + 118) 118)$
 $= 489.0655 \text{ kgf/cm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 118 mm, the minimum Shell length adjacent to the tubesheet should be 759.805 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 729^2 / ((1510 + 118) 118) [12.7 + (1.04^2 - 1)(195 - 150)] + 755^2 \cdot 150 / ((1510 + 118) 118)$
 $= 489.0655 \text{ kgf/cm}^2$

Shell Bending Stress due to Joint Interaction [σ_s, b]:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 2.45E+09 / 118^2 \{ 0.0041 [0.00211762 \cdot 195 + 755^2 \cdot 0 / (231731232)] + 6(1 - 0.32^2) / (745274) (729 / 280)^3 (1 + 280 \cdot 0.0041 / 2) [12.7 (0.16 + 0.68 \cdot -0.19) + 2 / 729^2 \cdot 0.68 \cdot -5963] \}$
 $= 1575.1582 \text{ kgf/cm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_{s, m}| + |\sigma_{s, b}| \leq 1.5 \cdot S_s$
 $:: |489| + |1575| \leq 2294 \text{ kgf/cm}^2$
 $:: 2064, \text{ must be } \leq 2294 \text{ kgf/cm}^2$

Note:
 For a given Channel thickness of 92 mm, the minimum Channel length adjacent to the tubesheet should be 670.896 mm

Channel Membrane Stress due to Joint Interaction [σ_c, m]:
 $= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$
 $= 755^2 \cdot 150 / ((1510 + 92) 92 \cdot \cos(0))$
 $= 579.764 \text{ kgf/cm}^2$

Channel Bending Stress due to Joint Interaction [σ_c, b]:
 $= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6(1 - \nu^2) / (E \cdot h) \cdot (a_o / h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 1370680064 / 92^2 \cdot \{ (0.002621014 \cdot 1 \cdot 0.0047 - 0.000 \cdot 1) \cdot 150 - 6(1 - 0.32^2) / 745274 \cdot (729 / 280)^3 \cdot (1 + 280 \cdot 0.0047 / 2 \cdot 1) \cdot [12.7 (0.16 + 0.68 \cdot -0.19) + 2 / 729^2 \cdot 0.68 \cdot -5963] \}$
 $= 2031.7794 \text{ kgf/cm}^2$

Channel Stress Summation vs. Allowable

$||\sigma_{c,m}| + |\sigma_{c,b}| \leq 1.5 \cdot S_c$
 $||580| + |2032| \leq 2465 \text{ kgf/cm}^2$
 $||2612| \text{ must be } \leq 2465 \text{ kgf/cm}^2 \text{ (See Elast-Plast iteration below)}$

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	252.7 <=	2110.7 kgf/cm ²	Pass
Tubesheet Shear Stress	34.2 <=	1125.7 kgf/cm ²	Pass
Maximum Tube Stress	9.7 <=	942.2 kgf/cm ²	Pass
Minimum Tube Stress (Buckling)	-9.7 <=	-366.1 kgf/cm ²	Pass
Maximum Force on any one Tube	14.9 <=	1443.7 kgf	Pass
Axial Membrane Stress in Shell	489.1 <=	1529.5 kgf/cm ²	Pass
Shell Stress (jt. inter.)	2064.2 <=	2294.3 kgf/cm ²	Pass
Channel Stress (jt. inter.)	2611.5 <=	2464.7 kgf/cm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:

= min(Sys, SPS,s/2)
 = min(2294, 4589/2) = 2294 kgf/cm²

Compute Allowable [S*c]:

= min(Syc, SPS,c/2)
 = min(2468, 4935/2) = 2468 kgf/cm²

Compute Factor [facts]:

= min([1.4 - 0.4* abs(σs,b) / Sstars], 1)
 = min([1.4 - 0.4* abs(1575)/2294], 1) = 1

Compute Factor [factc]:

= min([1.4 - 0.4* abs(σc,b) / Sstarc], 1)
 = min([1.4 - 0.4* abs(2032)/2468], 1) = 1

Note: Since facts & factc are both >= 1, the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	2064.2 <=	2294.3 kgf/cm ²	Pass
Channel Stress (jt. inter.)	2611.5 <=	4935.4 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	16.671	280.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	2.413	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	2.413	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc 240.404	...	1615	2111	362 1126	Ps+Pt-Th D1 Ok
D2uc 260.610	...	1867	2111	396 1126	Ps+Pt-Th D2 Ok



D3uc	16.671	...	253	2111	34	1126	Ps+Pt-Th	D3	Ok
O1uc	149.544	...	1649	4344	371	1126	Ps+Pt+Th	O1	Ok
O2uc	158.861	...	1833	4344	388	1126	Ps+Pt+Th	O2	Ok
O3uc	3.810	...	219	4344	25	1126	Ps+Pt+Th	O3	Ok
O4uc	3.810	...	34	4344	9	1126	Ps+Pt+Th	O4	Ok
D1c	250.874	...	1726	2111	375	1126	Ps+Pt-Th-c	D1	Ok
D2c	270.978	...	1992	2111	409	1126	Ps+Pt-Th-c	D2	Ok
D3c	27.969	...	265	2111	34	1126	Ps+Pt-Th-c	D3	Ok
O1c	160.599	...	1763	4344	384	1126	Ps+Pt+Th-c	O1	Ok
O2c	169.840	...	1955	4344	400	1126	Ps+Pt+Th-c	O2	Ok
O3c	15.810	...	229	4344	25	1126	Ps+Pt+Th-c	O3	Ok
O4c	15.810	...	36	4344	9	1126	Ps+Pt+Th-c	O4	Ok

Max: 270.9780 ... mm 0.944 0.364 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				:	Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd		
D1uc	42	1529	...	...		...	...	...	...		Ok
D2uc	447	1529	...	...		...	...	...	...		Ok
D3uc	489	1529	...	...		...	...	...	...		Ok
O1uc	33	4588	...	...		...	...	...	...		Ok
O2uc	438	4588	...	...		...	...	...	...		Ok
O3uc	480	4588	...	...		...	...	...	...		Ok
O4uc	9	4588	-9	-1788		...	...	...	...		Ok
D1c	47	1529	...	...		...	...	...	...		Ok
D2c	457	1529	...	...		...	...	...	...		Ok
D3c	504	1529	...	...		...	...	...	...		Ok
O1c	38	4588	...	...		...	...	...	...		Ok
O2c	448	4588	...	...		...	...	...	...		Ok
O3c	495	4588	...	...		...	...	...	...		Ok
O4c	9	4588	-9	-1786		...	...	...	...		Ok
Max RATIO	0.330			0.005		...			...		

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	141	942	-141	-366	217	1443	426	2294	2759	4935	Ok
D2uc	134	942	-116	-366	205	1443	2405	4588	147	2464	Ok
D3uc	10	942	-10	-366	15	1443	2064	2294	2612	4935	Ok
O1uc	143	1884	-123	-366	218	2887	441	4588	2782	4935	Ok
O2uc	152	1884	-89	-366	233	2887	2372	4588	123	4935	Ok

O3uc	27	1884	...	...	41	2887	2031	4588	2635	4935	Ok
O4uc	26	1884	...	...	41	2887	34	4588	24	4935	Ok
D1c	170	942	-170	-366	261	1443	490	2294	2910	4935	Ok
D2c	166	942	-116	-366	255	1443	2520	4588	191	2464	Ok
D3c	7	942	-7	-366	10	1443	2124	2294	2719	4935	Ok
O1c	153	1884	-153	-366	234	2887	507	4588	2935	4935	Ok
O2c	184	1884	-89	-366	282	2887	2484	4588	165	4935	Ok
O3c	30	1884	...	...	46	2887	2088	4588	2744	4935	Ok
O4c	26	1884	...	...	40	2887	35	4588	25	4935	Ok
Max RATIO	0.181	0.466	0.181	0.926	0.595						

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	270.978	280.000	Ok
Tube thickness :	2.112	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	2.413	Ok
Tube-tubesheet groove weld leg :	1.829	2.413	Ok

Min Shell length of thk, (118.000) adj. to tubesheet: 759.805 mm

Min Channel length of thk, (92.000) adj. to tubesheet: 670.896 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -0.123 mm
 Due to Pressure Shell Compresses by : -0.619 mm
 Due to Pressure + Thermal Shell Compresses by : -0.743 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	183.637	1.000	206.414	1.000
Tubesheet Shear Stress	446.577	1.000	535.680	1.000
Tube Tensile Stress	828.615	1.000	1102.79	1.000
Tube Compressive Stress	321.999	1.000	615.074	1.000
Tube-Tubesheet Joint load	828.617	1.000	1102.78	1.000
Shell Stress (Axial, Junction)	183.637	1.000	216.004	0.609
Tubesheet-Channel Junction	183.637	1.000	318.835	0.044
Tube Pressure Stress	363.358	1.000	318.835	1.000
Minimum MAWP	183.637		206.414	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	196.161	0.999	220.239	1.000
Tubesheet Shear Stress	465.977	1.000	553.135	1.000
Tube Tensile Stress	996.815	1.000	1368.48	1.000
Tube Compressive Stress	389.687	1.000	618.699	1.000
Tube-Tubesheet Joint load	996.813	1.000	1368.48	1.000
Shell Stress (Axial, Junction)	196.161	0.999	220.239	0.508
Tubesheet-Channel Junction	196.161	0.999	220.239	0.061
Tube Pressure Stress	363.358	1.000	-0.001	1.000
Minimum MAPnc	196.161		-0.001	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load	Due to Pressure	Due to Thermal Exp.	Computed Delta Ms
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Case#	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0374	0.0375	...	...	...	...
2	-0.6181	-0.6192	...	...	...	...
3	-0.5807	-0.5817	...	...	...	...
4	...	...	...	...	...	...
5	0.0374	0.0375	-0.1233	-0.1235	...	...
6	-0.6181	-0.6192	-0.1233	-0.1235	...	...
7	-0.5807	-0.5817	-0.1233	-0.1235	...	...
8	...	...	-0.1233	-0.1235	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(280/4, \min(280, 118)) \\
 &= 118.000 \text{ mm}
 \end{aligned}$$

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\
 &= \max(280/4, \min(280, 92)) \\
 &= 92.000 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio:

$$\begin{aligned}
 &= \min(\text{pt/Tubeside MAPnc}, 1) \\
 &= \min(150/196, 1) \\
 &= 0.764
 \end{aligned}$$

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

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